TOWARD CRYPTO-EXCHANGE OVERSIGHT

YESHA YADAV[†]

Abstract

The cryptocurrency market has proven notoriously challenging to regulate. Even as "crypto-winter" has left ordinary savers wiped out, regulation has shown itself to be woefully under-protective to either mitigate risks ex ante or help cover the damage ex post. This Article makes the case for mandating that crypto-exchanges perform industry oversight as a key part of comprehensive public reform of exchanges themselves as well as of the cryptocurrency markets in which they play an anchoring role. It offers three rationales in support. Firstly, a public mandate for such "self-regulation" brings crypto-exchanges in line with traditional financial exchanges that have long supervised their market – and been forced to clean their own houses in order to do so. This public mandate to deliver private oversight is rationally rooted. By promising a safe venue, exchanges can more fully perform their core economic function of enabling fluid trading, drawing users, helping them transact on predictable terms, and building network effects as numbers grow in response to increasing efficiencies and integrity. This designation of exchange-as-regulator thus helps align the private self-interest of exchanges with the public good. Secondly, crypto-exchanges occupy a unique position from which to perform oversight. As a centralizing firm connected to an array of crypto-users, technologies and ventures, exchanges possess systemic perspective and economic power. Especially within an ecosystem comprised of more decentralized infrastructure, the highly centralized and connected set-up of most crypto-exchanges lowers their cost of delivering oversight and increases its disciplinary power. Finally, by bringing crypto-exchanges within the regulatory perimeter, a publicly-overseen mandate for self-regulation creates a cost on exchanges for failing to comply. It forces venues to create standards for safety and to re-structure their own governance and operational practices in response, placing them – and, by extension the market – on a surer, more organized footing. In concluding, this Article lays out a roadmap for regulation to introduce a public mandate for self-regulation in crypto-markets, setting out strategies to incentivize successful private oversight. While recognizing that self-regulation has serious shortcomings, especially given the risks of

[†] Milton R. Underwood Chair and Professor of Law, Associate Dean, Vanderbilt Law School. I am deeply grateful for insightful comments, ideas and encouragement in the preparation of this piece. For their invaluable time and perspectives, I am thankful to Dan Awrey, Jonathan Brogaard, Chris Brummer, Anna Gelpern, Yuliya Guseva, Andrew Hayashi, Claire Hill, Madhu Kalimipalli, Elizabeth Pollman, Andriy Shkilko, Dan Schwarcz, Morgan Ricks, Danny Sokol, Fred Tung, Rory van Loo, Dan Webber, Hans de Wulf, Pradeep Yadav, David Zaring and to participants at the Rutgers Fintech Collaboratory, the BYU Winter Deals Conference, University of Virginia Law School/USC Law School Conference on Digital Platforms, Vanderbilt Law School/University of Ghent Law School Law and Business Conference, National Stock Exchange of India's Workshop Series, Boston Law School Seminar on Financial Markets, University of Minnesota Faculty Workshop, Yale Law School/Georgetown Law School/Wharton Business School/Queen Mary University Law/Bank of England Conference on Law and Macroeconomics Comments, and the Wilfred Laurier Business School Faculty Workshop. Further comments are are most welcome at yesha.yadav@vanderbilt.edu. All errors are my own.

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crypto-exchanges, it views exchange oversight as one important part of wider regulatory reform, urgently needed to address the significance of the crypto-economy and its deeper entanglement within the financial system and in the lives of everyday savers.

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I. INTRODUCTION

When a QR-code floated across television screens during Superbowl 2022, those scanning it were led to the website for Coinbase – the second largest cryptocurrency exchange in the world by volume.[1] The advertisement was wildly popular, crashing the website and increasing downloads of Coinbase's mobile-app by 309% over the next week.[2] Coinbase was far from the only crypto-platform to make a grab for the Superbowl's prestige advertising slots.[3] Crypto.com, EToro, and FTX, then-fronted by A-List celebrities, all made prime-time debuts, and signaled the arrival of cryptocurrencies into the coveted echelons of mainstream Americana.[4]

By the close of 2022, however, the fate of cryptocurrency exchanges could hardly have looked bleaker. In November, FTX – commercially dominant and posing as the compliance-forward face of crypto – collapsed into a chaotic, complex bankruptcy, with co-founder and former CEO, Sam Bankman-Fried charged with multiple counts of criminal wrongdoing.[5] FTX's failure left its $32 billion valuation in tatters, and around nine million creditors across 134 entities facing lengthy, legal limbo in a bid to recover their money/crypto.[6] The FTX bankruptcy filing revealed a sprawling organization in disarray, lacking basic recordkeeping, corporate governance, and risk management – capable of losing eight billion dollars in customer funds in apparent self-dealing, recklessness and hubris.[7] A key pillar of the crypto-economy, FTX's bankruptcy filing had immediate ripple effects, forcing multibillion-dollar crypto-lenders, BlockFi and Genesis Global, into bankruptcies of their own.[8]

But FTX is just one of multiple leading crypto-exchanges to find itself in troubled waters. In 2023, the Securities and Exchange Commission (SEC) sued three major exchanges – Binance, the world's largest crypto-exchange by

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[1] COINMARKETCAP, TOP CRYPTOCURRENCY SPOT EXCHANGES, https://coinmarketcap.com/rankings/exchanges/ (last visited: Aug. 6, 2023) (measuring spot-trading); Anthony Tellez, *Crypto Ads Are a Super Bowl Talker, with Floating QR Codes and Larry David*, NPR, Feb. 13, 2022.

[2] Sarah Perez, *Super Bowl Ads Boosted Crypto App Downloads by 279%, Led by Coinbase*, TECH CRUNCH, Feb. 17, 2022.

[3] THE BLOCK CRYPTO, COMPANIES: COINBASE, https://www.theblockcrypto.com/data/crypto-markets/public-companies (detailing verified user numbers as at end-2021) (May 2022); Coinbase, *Shareholder Letter: Second Quarter 2023* (Aug. 3, 2023), 5-6.

[4] FTX and Crypto.com became named sponsors for sporting arenas. CRYPTO.COM ARENA, ARENA INFO, https://www.cryptoarena.com/about-us; FTX ARENA, ABOUT, https://www.ftxarena.com/about; Tiffany Hsu, *Prepare Yourself for This Weekend's 'Crypto Bowl,'* N.Y. TIMES, FEB. 13, 2022.

[5] *See e.g.*, Dan Mangan & Brian Schwartz, *FTX Founder Sam Bankman-Fried Hit with Four New Criminal Charges*, CNBC, Feb. 23, 2023.

[6] *See sources cited infra* note [306]; Noele Illien, Tom Wilson and Dietrich Knauth, *FTX Opposes New Bankruptcy Investigation As It Probes Bankman-Fried Connections,* REUTERS, JAN. 26, 2023; *Factbox: From BlockFi to Genesis, Crypto Firms Reel from Exposure to FTX*, REUTERS, JAN. 5, 2023.

[7] Joshua Oliver, Nikou Asgari & Kadhim Shubber, *FTX: Inside the Crypto Exchange that 'Accidentally' Lost $8bn*, FIN. TIMES, NOV. 18, 2022.

[8] Oliver, Asgari & Shubber, *supra* note [7].

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trading volume, Coinbase and Bittrex Inc, alleging that the venues were operating illegally as unregistered securities exchanges.[9] In March 2023, Binance was charged by the Commodity and Futures Trading Commission (CFTC) for suspected breaches of applicable laws, and is facing further scrutiny from federal regulators for possible violations of money laundering regulations.[10] January 2023 saw the SEC file a complaint against Gemini Trust for violations of securities laws.[11] In Fall 2022, Bittrex Inc. and Kraken entered into settlements with U.S. authorities for failures to prevent money laundering and illicit money flows.[12] This legal fallout has come with serious financial and reputational costs. In May 2023, for example, Bittrex Inc. filed for Chapter 11 bankruptcy, citing pressures of regulatory action as a reason for shuttering its U.S. operations.[13] Binance has experienced billions in customer outflows as crises and regulatory problems have mounted – and lost a third of its market share in just three months between February-June 2023.[14] Coinbase has seen sharply lower trading volume in 2023 despite rising crypto-prices.[15]

These whipsawing fortunes highlight two factors significant to the oversight of crypto-exchanges and that of the broader market. First, exchanges represent critical, centralizing pillars within the crypto-ecosystem. They convene a diverse cast of stakeholders within their institution. Offering everyday customers an easy way to transact in cryptocurrencies, exchanges have enabled the widespread popularization of an otherwise technical and relatively novel asset class. Combining crypto-trading with the offer of lucrative financial products, social media savvy and catchy marketing, top exchanges have, despite their troubles, acquired expansive user bases and become repositories for billions of dollars in customer assets. Additionally, by listing tokens, continually including new digital assets for trading, major exchanges constitute crypto's key gatekeepers, determining which coins can enjoy marketplace visibility and a chance at economic viability. In the historic absence of a public regulator like the SEC that oversees the entry of new public

[9] Sec.& Exch. Comm'n, *SEC Files 13 Charges Against Binance Entities and Founder Changpeng Zhao*, Press Release, Jun. 5, 2023; Sec.& Exch. Comm'n, *SEC Charges Crypto Asset Trading Platform Bittrex and its Former CEO for Operating an Unregistered Exchange, Broker, and Clearing Agency*, Press Release, Apr. 17, 2023; Sec.& Exch. Comm'n, *SEC Charges Coinbase for Operating as an Unregistered Securities Exchange, Broker, and Clearing Agency*, Press Release, Jun. 6, 2023.

[10] Elizabeth Napolitano, *US Investigators Subpoena Hedge Funds in Binance Money-Laundering Probe: Report*, COINDESK, JAN. 7, 2023; Commodity and Futures Trading Comm'n, *CFTC Charges Binance and Its Founder, Changpeng Zhao, with Willful Evasion of Federal Law and Operating an Illegal Digital Asset Derivatives Exchange*, Press Release, Mar. 27, 2023.

[11] Ephrat Livni, *S.E.C. Charges Crypto Companies With Offering Unregistered Securities,* N.Y. TIMES, JAN. 12, 2022.

[12] Menqi Sun, *Crypto Exchange Kraken Settles Alleged Sanctions Violations*, WALL ST. J., NOV. 29, 2022.

[13] Daniel Lowenthal & Kimberly Black, *The Unique Case of Bittrex*, Patterson Belknap Bankruptcy Update (May 24, 2023).

[14] Javier Paz, *Binance is Bleeding Assets, 12 Billion Gone in Less Than 60 Days*, FORBES, JAN. 10, 2023; Vicky Ge Huang & Dave Michaels, *Binance Sees $2 Billion in Outflows as Troubles Compound*, WALL ST. J, MAR. 29, 2023; Scott Chipolina, *Binance Loses Market Share after Regulatory Clampdown*, FIN. TIMES. JUN. 2, 2023.

[15] Scott Chipolina, *Coinbase Trading Volumes Fall as Regulatory Uncertainty Weighs*, FIN. TIMES. APR. 23, 2023.

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companies into the market, crypto-exchange decision-making has taken center-stage, vetting new digital assets (and the ventures that back them), and shaping the risks to which customers are exposed. Finally, the economic centrality of many cryptocurrency exchanges has encouraged industry reliance on their services and financial firepower. As revealed by FTX, major crypto-exchanges can hold billions in assets, use their balance sheets to lend and borrow with other firms, and sometimes take equity stakes in new and established ventures. FTX's corporate group, when it failed, was holding a portfolio of around 500 equity investments, worth around $5.4 billion, including stakes in companies like Elon Musk's Space X, Sequoia Capital, and the Investors Exchange (IEX), a regulated national securities exchange.[16] The market's economic dependence on the power of crypto-venues underscores the general risk to other institutions from exchange failure, through the interconnections an exchange's financial relationships can create between disparate parts of the crypto-ecosystem.[17]

Secondly, despite their significance, sophistication, and financial footprint, crypto-exchanges have evolved business models, governance and trading practices largely outside the purview of a dedicated framework for regulation and supervision. Even as committees in Congress have periodically introduced bills for oversight, the challenge of devising and executing a plan in practice has become clear by the persistence of deep-seated differences between agencies on foundational questions concerning turf and jurisdiction.[18] Uncertainty about legal classifications for cryptocurrencies – whether these might be categorized as securities or commodities or something else entirely, and which agency ought then to be in charge – has ultimately stymied efforts to bring crypto-exchanges into the federal regulatory perimeter.[19] In the absence

[16] Kadhim Shubber and Bryce Elder, *Revealed: The Alameda Venture Capital Portfolio*, FIN. TIMES (ALPHAVILLE), DEC. 6, 2022.

[17] On considerations of systemic risks in the cryptocurrency market more broadly, see e.g., Observing systemic fragility in decentralized lending markets, Alfred Lehar & Christine Parlour, *Systemic Fragility in Decentralized Markets*, Working Paper (Jul. 2022); Aaron Klein, *The Future of Crypto Regulation: Highlights from the Brookings Event*, BROOKINGS, AUG. 11, 2022 (detailing discussions on the high level of retail participation and vulnerability; Owen Lock and Teresa Cascino, *Cryptoassets, the Metaverse and Systemic Risk*, BANK OF ENGLAND UNDERGROUND, AUG. 9, 2022, https://bankunderground.co.uk/2022/08/09/cryptoassets-the-metaverse-and-systemic-risk/ (highlighting the potential for systemic risks within crypto-markets).

[18] *See for example*, Jesse Hamilton, Don't Pop the Champagne on U.S. Crypto Bills – Progress in Congress Has Been Costly, COINDESK, AUG. 2, 2023; See also, the White House, *Executive Order on Ensuring Responsible Development of Digital Assets*, March 9, 2022, https://www.whitehouse.gov/briefing-room/presidential-actions/2022/03/09/executive-order-on-ensuring-responsible-development-of-digital-assets/ (emphasizing the need for research to better understand the economic and legal issues surrounding digital assets); PRESIDENT'S WORKING GROUP ON FINANCIAL MARKETS, REPORT ON STABLECOINS, 1, 11 (NOV. 2021) (highlighting the potential authority of the SEC and the CFTC over certain stablecoin-type of assets and the challenges in determining jurisdiction).

[19] State-based regimes offer more comprehensive options, specifically in New York and Wyoming. For discussion see sources *infra* note [267]. Further, discrete functional regulation may be triggered for those transacting in derivatives or making payments (engaging anti-money laundering), for example. Klein, *supra* note [17]; James Nani, *Crypto Winter Triggers Calls for Clear Bankruptcy Guidelines*, BLOOMBERG LAW, JUL. 15, 2022. On the challenges of regulating new technologies, John Bagby & Nizan Packin, *RegTech and Predictive Lawmaking: Closing the RegLag between Prospective Regulated Activity and Regulation*, Working Paper (Aug. 2020) (detailing "regulatory lag" and "legal lag" where rules to oversee a regulatable activity take time to create).

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of systematic monitoring and rulemaking, policymakers have been caught off-guard by the centrality, sophistication, and underlying fragilities of cryptocurrency exchanges, without levers to either control the risks *ex ante* or to respond to fallout *ex post*.[20] Regulatory gaps have proven costly. For example, exchange customers have lacked clarity about the legal quality of the assets they hold on a crypto-exchange, with the result that users risk ending up as unsecured creditors of an insolvent venue rather than recognized as owners of their crypto and entitled to immediate and full repayment.[21]

To address the risks posed by crypto-exchanges, alongside the need for public oversight, this Article makes the case for mandating private self-regulation by crypto-exchanges as a first step within the larger regulatory agenda for digital assets.[22] That is, regulation would mandate that crypto-exchanges craft credible standards and rules for the industry, monitor and discipline those using their venues, support public regulators in providing crypto-market integrity. Importantly, under the mandate, platforms would have to prove to regulators that they are up to the job – or face restrictions in their ability to offer a full suite of trading and other financial services to customers. To do so, crypto-exchanges would be tasked with demonstrating strong internal corporate governance, risk management, limited conflicts of interest, and credible mechanisms to protect customer assets. By virtue of this mandate, exchanges would come under a public supervisory regime as a key function of their digital asset business. They would face accountability for their failures to act as stewards of the market and incur a cost on their conduct that they do not pay presently. To be clear, private self-regulation is not a substitute for a comprehensive, detailed public oversight framework for cryptocurrencies. But

[20] On conceptual challenges, see e.g., Chris Brummer, *Disclosure, Dapps & De-Fi*, Working Paper, 2-4 (Mar. 24, 2022) (noting the difficulties of applying traditional securities law paradigms, notably disclosure, to decentralized finance); Chris Brummer, Jai Massari & Trevor Kiviat, *What Should be Disclosed in an ICO?* CRYPTO-ASSETS: LEGAL, REGULATORY & MONETARY PERSPECTIVES, 157-202 (2019); Chris Giancarlo & Justin Browder, *Foundational Principles for US Crypto Asset Regulation*, IFLR, FEB. 17, 2022 (pointing to *ad hoc* nature of regulatory efforts). Decentralized finance – where protocols operate on smart contracts have also proliferated and developing increasing sophistication and financialization, On sophistication and financialization in decentralized finance, see, Lin William Cong & Zhiguo He, *Blockchain Disruption and Smart Contracts*, NBER Working Paper No. w24399 (March 2022) (detailing complex contracting within blockchains); Fabian Schär, *Decentralized Finance: On Blockchain and Smart Contract-Based Financial Markets*, Fed. Res. Bank of St. Louis Rev., 153(2021) (detailing various applications of smart contracts and defi protocols).

[21] Adam Levitin, *What Happens if a Crypto Exchange Files for Bankruptcy*, CREDIT SLIPS, (Feb. 4, 2022), https://www.creditslips.org/creditslips/2022/02/what-happens-if-a-cryptocurrency-exchange-files-for-bankruptcy.html.

[22] On the benefits of self-regulation for banking and financial markets, see generally, Saule Omarova, *Wall Street as a Community of Fate: Toward Financial Industry Self-Regulation,* 159 U. PENN. L. REV. 411 (2011); Cary Coglianese & Evan Mendelsohn, *Meta-Regulation & Self-Regulation*, U. of Penn Law School, Public Law Research Paper No. 12-11 (2011) (noting the discretion afforded to self-regulating firms). In July 2023, Senators Lummis and Gillibrand reintroduced their Bill for outlining a comprehensive set of reforms for digital assets – The Responsible Financial Innovation Act – which newly contains draft provisions designed to make crypto-exchanges self-regulatory organizations or SROs. Draft section 404 of the Bill sets out a framework for exchange self-regulation. Lummis, Gillibrand Reintroduce Comprehensive Legislation To Create Regulatory Framework For Crypto Assets, Press Release, Jul. 12, 2023, https://www.gillibrand.senate.gov/news/press/release/lummis-gillibrand-reintroduce-comprehensive-legislation-to-create-regulatory-framework-for-crypto-assets/.

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it offers a starting point that can be deployed to engineer a systematic compliance focus in the short-term, while offering an anchor for public regulators as part of a maturing digital asset industry in the longer-term.

This Article outlines three reasons in support of this argument. One, mandatory exchange self-regulation brings crypto-markets in line with the broader financial system, that relies heavily on exchanges for oversight.[23] As detailed in Part II, exchanges have a long history of organizing and policing their markets. From the Republic's early days, self-regulation has backstopped capital markets by establishing a framework for standardizing traded claims, the mechanisms for their trading, and deciding who gets to use their venues.[24] Exchanges like the New York Stock Exchange or Nasdaq have thus wielded expansive economic and reputational power to generate conformity with industry rules, standards, and norms.[25] By credibly threatening users with the prospect of sanction and expulsion, exchanges incentivize good behavior from those that need access to the platform's facilities.[26]

Part II describes why exchanges, in particular, have come to play this regulatory role to show that platforms can gain by investing in good governance. Order and standardization help exchanges to perform their core economic function as intermediaries seeking to produce a vibrant marketplace for claims. As Paul Mahoney details, by creating common rules and contract terms, understood and agreed-upon by exchange participants, trading can proceed without requiring parties to bargain, monitor performance, and litigate breaches and bad behavior from counterparties.[27] Parties can rely on the expertise and power of the exchange instead.[28] Trust, confidence and reliability help exchanges attract more users, encouraging network effects where increasing numbers of customers build the capacity of the exchange to match buyers and sellers, reducing trading costs, and producing richer information about assets.[29] As with traditional venues, crypto-exchanges too can benefit

[23] Omarova, *supra* note [22], 431-438; Kimberly Krawiec, *Return of the Rogue*, 159 AZ. L. REV. 127 (2009) (examining "enforced self-regulation" in banking regulation and detailing the risks of introducing such a regime in risk management); Jerry Markham & Daniel Harty, *For Whom the Bell Tolls: The Demise of Exchange Trading Floors and the Growth of ECNs*, 33 J. CORP. L. 865, 867-80 (2008) (detailing the historic rise of exchanges as key regulators in securities markets).

[24] Markham & Harty, *supra* note [23], 867-870.

[25] *See, e.g.*, Exchange Act § 6(a), 15 U.S.C. § 78f(b) (2000) (stipulating that national exchanges must ensure standards of fairness and integrity on their venues); Commodity Exchange Act, 7 U.S.C. §§ 1–27 (2006) (on self-regulation by commodities and derivatives exchanges); SECURITIES & EXCH. COMM., CONCEPT RELEASE CONCERNING SELF-REGULATION, 69 Fed. Reg. 71256 (proposed Dec. 8, 2004), available at http://sec.gov/rules/concept/34-50700.pdf (noting the significance of self-regulation to the regulatory system for securities markets); For an international perspective, Stavros Gadinis & Howell E. Jackson, *Markets as Regulators*, 80 S. CAL. L. REV. 1239, 1244 (2007) (highlighting exchange-based regulation comparatively).

[26] For an overview, see, Paul Mahoney, *The Exchange as Regulator*, 83 VA. L. REV. 1453, 1459-1462 (1997) (detailing the Buttonwood Agreement and the reasons for exchange self-regulation).

[27] Mahoney, *supra* note [26], 1459-1461.

[28] For discussion of advantages of self-regulation generally, including values of flexibility and speed, Omarova, *supra* note [22], 422-424; Danny Sokol & Roisin Comerford, *Antitrust & Regulating Big Data*, 23 GEORGE MASON L. REV. 1129, 1136-1137 (2016)(detailing network effects and their creation).

[29] *For discussion see infra* Part [II. A].

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privately from producing a lively and liquid marketplace of buyers and sellers, capable of producing a loyal customer base with the promise of safety and honesty. This private incentive can produce public gain, where it motivates exchanges to each competitively invest in oversight and good governance, with the result that market integrity is enhanced at an aggregate level.[30]

Two, exchanges occupy a unique position within the crypto-economy, convening a variety of diverse stakeholders under one roof. This can reduce their costs of providing monitoring and discipline, such that they should arguably constitute the least cost provider of supervision across the market.[31] As a starting point, crypto-exchanges draw together retail as well as institutional users who can transact in an environment that does not require engaging with more decentralized and technically daunting "blockchains."[32] This centralization contrasts with the philosophical vision underlying cryptocurrency's origins, but it comes with advantages for users that can avoid the need for technical learning, navigating delays and internalizing shifting fee structures that are typical across major public blockchains.

By way of summary background, Bitcoin's core concept seeks to circumvent the need for the kind of conventional intermediation exchanges provide.[33] Rather than use platforms, banks, or governments to police transfers of value, individual computers – applying code in a self-executing, automated way – complete the necessary checks entailed in enabling value to move between users. Computers, together, run the code and check the validity of transactions, accepting transfers as executed only once their terms have gained sufficient consensus within the network (e.g., 51%). Once accepted by a threshold number of participant "nodes," verified trades form part of the network's record of accepted transactions – colloquially, a distributed ledger or blockchain – that records and publicly displays details of trades.[34]

Their intriguing design aside, transacting on public blockchains entails costs, especially for a newcomer public. Finding a counterparty is challenging in an environment where identities are obscured behind pseudonymous handles.[35] Busy blockchains can be slow to accept and finalize transactions –

[30] For example, in announcing its partnership with BlackRock, Coinbase repeatedly mentions safety and compliance as a key feature. Coinbase, *Coinbase Selected by BlackRock; Provide Aladdin Clients Access to Crypto Trading and Custody via Coinbase Prime*, Press Release, Aug. 4, 2022.

[31] Some crypto-exchanges operate as "decentralized" venues run on blockchain-based smart-contracts. This Article does not focus on decentralized venues. For analysis, Alfred Lehar & Christine Parlour, *Decentralized Exchanges*, Working Paper (2022)(highlighting gains of decentralized exchanges). In tort law, Professor Guido Calabresi, highlighted the gains of attributing liability to the actors that can avoid creating liability at lowest cost – that is, the party that is best situated to control the harms arising at least cost. This least cost avoider principle offers a lens to consider which actor may be able to offer supervision at least cost, increasing the efficiency of monitoring within the oversight framework. On tort law's cheapest cost avoider, GUIDO CALABRESI, THE COST OF ACCIDENTS: A LEGAL AND ECONOMIC ANALYSIS (1970).

[32] Coinbase, *Shareholder Letter*, *supra* note [3], 5-6 (noting that institutional traders generated around $78 billion of the $92 billion dollars of trading volume on the exchange in Q2, 2023.

[33] For discussion on the problematic use of the term decentralization, Walch, *infra* note [188].

[34] *See discussion and sources infra* Part [III. A].

[35] *See discussion and sources infra* Part [III.].

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and some may fail to be processed. User fees can also become prohibitively high where surging volumes result in higher charges.[36] Importantly, by emphasizing disintermediation and decentralization, parties lack recourse in the event of mistake or misconduct. Within an environment of self-sufficiency and the need for users to acquire technical competency, a blockchain-only trading environment inhibits a mass-market and limits traditional pathways for oversight and discipline to be exercised cheaply and effectively.[37]

Crypto-exchanges, by contrast, provide a familiar mechanism for trading outside of blockchains, presenting users with a known form of central intermediary. Instead of the search costs on blockchains, exchanges connect buyers and sellers within an organized trading environment.[38] Rather than finalize trades on blockchains, exchanges mostly settle them in-house, moving value between user accounts on the books of the venue itself.[39] This enables trades to be settled rapidly, rather than requiring customers to wait to receive confirmation that the transfer has been recognized on a blockchain.[40] By consolidating identified users within their platform, and typically having involvement in virtually all aspects of the trading process (e.g., marketing, order matching, settlement, custody), exchanges are singularly positioned to monitor user activities, trading habits, and potential for misconduct.[41]

In addition to hosting traders on the venue, exchanges also list various digital assets connected to different blockchains and entrepreneurial ventures. That is, beyond allowing trades in quintessential cryptocurrencies like Bitcoin and Ether, exchanges routinely list other types of digital assets, including those reflecting interests in crypto-ventures that themselves seek to operate in a decentralized manner, using automated code to verify and execute financial transactions like lending or insurance. By listing coins native to any number of ventures and blockchains, including those looking to operate in a decentralized way, exchanges arguably gain an interest in ensuring the basic viability of these enterprises and infrastructures. That is, by gaining exposure to users, technologies, and the risk of ventures and blockchains failing, exchanges acquire skin-in-the-game to better ensure that that these stakeholders maintain economic, legal and technological viability.[42]

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[36] *See discussion and sources infra* Part [III. AIII. A].

[37] *See discussion and sources infra* Part [III. A]. To be sure, regulators and industry participants are exploring mechanisms to embed automated regulatory features into decentralized blockchains, however, work remains in progress. For discussion of the concept, Raphael Auer, *Embedded Supervision: How to Build Regulation into Decentralized Finance,* BIS Working Papers Number 811 (May 2022).

[38] *See discussion and sources infra* Part [III. B].

[39] *See discussion and sources infra* Part [III. B].

[40] *See discussion and sources infra* Part [III.]. See e.g., Aoyon Ashraf, *Crypto Exchange Coinbase to Offer Liquid Staking Token Before Ethereum Merge,* COINDESK, AUG. 25, 2022.

[41] For discussion of the centrality of crypto-exchanges and explaining why they have evolved as one-stop-shops in crypto market structure see, for example, Yesha Yadav, *The Centralization Paradox in Cryptocurrency Markets*, 100 WASH. U. L. REV. 1725(2023).

[42] *See discussion and sources infra* Part [III.].

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Their centrality to an otherwise diffuse and diverse crypto-ecosystem elevates exchanges to a position from which they are uniquely capable of performing oversight. Moreover, relative economic importance (e.g., in listing new tokens) reinforces their power, bolstering the coercive weight of rulemaking and private discipline. By combining position, power and stature, crypto-exchanges present policymakers with a lower-cost, high-coverage monitor that can offer a regulators a supportive complement to enhance industry surveillance and market integrity across an asset class otherwise designed to operate in a manner impervious to a single government oversight and control.[43]

Three, mandating self-regulation by crypto-exchanges constitutes a mechanism by which policy can compel exchanges to follow standards of good governance and institutional prudence. As demonstrated most visibly by the collapse of FTX, alongside the swirl of regulatory suspicion regularly surrounding leading crypto-exchanges, the absence of regulatory coverage and clarity has resulted in the creation of organizational structures that can pose serious risks to customers, and to the crypto-market itself. A mandate for self-regulation brings crypto-venues firmly into the regulatory perimeter. In traditional markets, the requirement for financial exchanges to discharge a oversight means that exchanges have to demonstrate organizational capacity and skill to perform this role.[44] Breaches can attract hefty fines and public fallout.[45] Such demands constrain the ability and incentive of exchanges to engage in unfair or excessively self-interested behavior (e.g., promoting the interests of their most favored traders).[46] It also forces them into regular conversation with public regulators to aid in furthering public policy goals through exchanges' own (usually more flexible) abilities for rulemaking and surveillance.[47] By subject venues to a mandate for self-regulation – and a cost when they fall short – public regulation helps reduce the odds of individual exchanges defecting from a public goal in pursuit of their own private ends. Part IV explores ideas for formally incentivizing exchanges to invest meaningfully in good governance. Traditional venues, for example, are rewarded with benefits like qualified immunity from lawsuits.[48] Crypto-exchanges might seek out similar protections. Designated crypto-venues might

[43] *See e.g.*, Douglas Michael, *Federal Agency Use of Audited Self-Regulation as Regulatory Technique*, 47 ADMIN. L. REV. 171, 180-185 (1995)(detailing the benefits, such as expertise, for regulators and showing the reduced costs faced by taxpayers).

[44] *See sources and discussion infra* notes [134]-[139]. Former Chair of the CFTC, Tim Massad, has proposed setting up an oversight board comprising the SEC and the CFTC to oversee the cryptocurrency market, Fran Velasquez, *Former CFTC Chair: Here's How SEC and CFTC Can Work Together to Regulate Crypto*, COINDESK, AUG. 22, 2022.

[45] *See sources and discussion infra* notes [134]-[139]. There are certainly arguments that regulators ought to demand more intensive exchange oversight and ought themselves to be wary of capture to hold industry accountable. For discussion, *See* Andrew Tuch, *The Self-Regulation of Investment Bankers*, 83 GEO. W. L. REV. 101, 137–40 (2015) (noting concerns about the small number of actions brought); Omarova, *supra* note [22], 463-464.

[46] *See sources infra* notes [135]-[136].

[47] *See e.g.*, Thompson, *infra* note [96]; Omarova, *supra* note [22], 423.

[48] City of Providence v. BATS Global Markets, Inc., 878 F.3d 36 (2d Cir. 2017).

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gain by enjoying wider commercial opportunities permitted to those that can show a more credible organizational and oversight-ready structure. Finally, where leading exchanges all internalize compliance costs to switch to a self-regulatory model, the market as a whole takes concrete steps to recover lost trust. Tarnished by the collapse of FTX and other crypto-businesses, a formal commitment to exchange self-governance can constitute a first step toward recovering public confidence, attracting customers, securing network effects that help building legitimacy for the crypto-industry more widely.

To be sure, this Article is realistic about the power of exchange self-governance to oversee a marketplace. This model is imperfect, as is well-established in the literature. As Jonathan Macey and Maureen O'Hara write, traditional exchanges are subject to powerful commercial interests that can feed deep conflicts of interest in oversight.[49] On a broader level, scholars have critiqued the very notion of exchange-based self-regulation as a viable source of public protection. Trusting industry to serve as its own security raises obvious worries that private firms will find ways to enrich themselves, be overly lax, or prevent top-down rulemaking by emphasizing their own supervision as being sufficient.[50]

But, exchange-led self-regulation cannot be the only solution to limited comprehensive oversight in crypto-markets. Rather, it assumes that public regulation will take center-stage – first, to implement a mandate, and eventually to set-up a detailed public supervisory framework. Furthermore, as Saule Omarova writes, self-regulation does not equal deregulation – and certainly does not in this proposal. As discussed in Part III, and evidenced by their 2022 crisis, cryptocurrency markets clearly pose risks that must be understood and mitigated. The costs of their collapse have been extensive and painful, particularly by relative newcomers to markets, including underrepresented minorities and younger Americans, many of whom have remained excluded in traditional financial markets but who have nevertheless seen opportunities for wealth-building in crypto.[51] Taken as a whole, the aim of this Article lies in taking a first step toward developing a market-wide framework for reform, one whose need has become urgent in the relative absence of public regulation. It relies on a tried-and-tested model of oversight within traditional markets, designed to to encourage the crypto-industry to take responsibility for looking after their own market in a bid to place it and themselves on a safer footing.

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[49] Jonathan Macey & Maureen O'Hara, *From Markets to Venues: Securities Regulation in an Evolving World*, 58 STAN. L. REV. 563 (2005).

[50] Omarova, *supra* note [22], 423; Julia Black, *Decentring Regulation: Understanding the Role of Regulation and Self-Regulation in a 'Post-Regulatory' World*, 54 CURRENT LEG. PROBLEMS 103, 114-116 (2001).

[51] Chris Brummer, *Congressional Testimony on the Future of Digital Assets Regulation Before the Agriculture Committee, U.S. House of Representatives Subcommittee on Commodity Exchanges, Energy, and Credit,* Jun. 23, 2022; Klein, *supra* note [17]. For example, an October 2021 survey found that 23 percent of Black Americans and 17 percent of Hispanic Americans owned cryptocurrencies, compared to 11 percent of white Americans – showcasing popularity rend that appears even more pervasive among younger people. Yahoo, *African-Americans Outpace Whites and Hispanics In Cryptocurrency Investments, Is a New Avenue to Black Wealth-Building Opening?*, Oct. 26, 2021.

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II. THE ROLE OF EXCHANGES IN OVERSIGHT

Securities and commodities exchanges have become critical providers of regulation within the U.S. supervisory system.[52] Capital markets have grown up under a model in which oversight by exchanges is the norm – if not often the most direct form of supervision.[53] This Part details why exchanges have become go-to gatekeepers. Designated as self-regulatory organizations (SROs), they wield expansive power across the economy to write the rules-of-the-road, monitor and enforce their implementation, and punish misbehavior.[54]

To be sure, exchange oversight comes with problems like conflicts of interest that arise because venues regulate firms that constitute their customers.[55] These shortcomings aside, however, the concept of exchanges serving the public interest has become entrenched within regulation, reflecting a public-private bargain struck in the provision of oversight.[56]

A. Why Exchanges Regulate Financial Markets

Exchanges have been present within the oversight structure of financial markets since almost the first days of the Republic.[57] In the early 1790s, New York stockbrokers laid out the conditions on which they would trade assets,

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[52] Chris Brummer, *Stock Exchanges and the New Markets for Securities Laws*, 75 U. CHI. L. REV. 1435, 1452 (2008) (arguing exchanges to be sellers of "law"); Stavros Gadinis & Howell E. Jackson, *Markets as Regulators*, 80 S. CAL. L. REV. 1239, 1244 (2007) (highlighting exchange-based self-regulation internationally).

[53] For example, companies that list on exchanges – the New York Stock Exchange or Nasdaq – must comply with a detailed set of listing rules on granular aspects their workings (e.g., composition of corporate governance committees). See for example, NYSE, LISTED COMPANY MANUAL, RULE 303A: CORPORATE GOVERNANCE STANDARDS, https://nyseguide.srorules.com/listed-company-manual/document?treeNodeId=csh-da-filter!WKUS-TAL-DOCS-PHC-%7B0588BF4A-D3B5-4B91-94EA-BE9F17057DF0%7D--WKUS_TAL_5667%23teid-68.

[54] *See, e.g.*, Exchange Act §6, 15 U.S.C. § 78f(b) (2000) (requiring national exchanges to maintain marketplaces that are free of fraud and manipulation and maintain standards of equitable and fair conduct).

[55] *See e.g.*, City of Providence v. BATS Global Markets, Inc., 878 F.3d 36 (2d Cir. 2017), 41-48 (describing how exchanges offer a range of paid products to enable high-speed trading, such as rapid data feeds and the ability for traders to locate their servers next to those of the exchange). For discussion, Yesha Yadav, *Insider Trading and Market Structure*, 63 UCLA L. REV. 968 (2016) (detailing the profitable nature of data-provision and colocation services that exchanges provide); Macey & O'Hara, *supra* note [49].

[56] SECURITIES & EXCH. COMM., CONCEPT RELEASE CONCERNING SELF-REGULATION, 69 Fed. Reg. 71256 (proposed Dec. 8, 2004), available at http://sec.gov/rules/concept/34-50700.pdf; See for example, N.Y. Stock Exch. LLC, Exchange Act Release No. 34-72065, 108 SEC Docket 3659, 2014 WL 1712113, 2-3 (May 1, 2014) (noting the role of exchanges as providers of rules); Weissman v. NASD, Inc. (Weissman IV), 500 F.3d 1293, 1299 (11th Cir. 2007) (distinguishing the roles that exchanges play as commercial versus regulatory entities).

[57] Paul Mahoney, *The Exchange as Regulator*, 83 VA. L. REV. 1453, 1459-1462 (1997); Joel Seligman, THE TRANSFORMATION OF WALL STREET, 73-76 (Aspen 3rd Ed., 2003) (detailing the origins of modern exchanges from private agreements between brokers).

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privately regulating issues like when contracts became binding, how they would be performed (e.g., delivery times), the consequences of breach and so on.[58] These collective norms led to advantages for the market. Arms-length contract parties saved themselves the cost of negotiating new terms for every transaction.[59] Instead, they were bound to rules already agreed-to by the group.[60] Market participants could better count on bargains being honored and rely on peer discipline to ensure that counterparties could be trusted.[61] Contracts made between exchange members signaled that participants had the resources to follow through on their promises – and to pay up if they broke them.[62] Rules requiring members to prove their solvency worked to create repeat players that could better absorb the risk of their bargains. Those that defaulted were censured or expelled.[63] Importantly, shared norms helped create confidence that the market was safe. Exchange rules looked to reduce fraud, manipulation and sought to verify that claims being traded were legally sound.[64] As Paul Mahoney writes, private oversight provided the first burst of organization for public markets, predating the intervention of public regulators like the Securities and Exchange Commission (SEC) by more than a century.[65]

In other words, private oversight supports the key economic function of exchanges.[66] Transacting fluidly demands norms that motivate buyers and sellers to interact in ways that transfer claims efficiently, produce informative prices, and encourage capital to be allocated productively.[67] Exchanges that succeed can encourage further business to come to their venue as more traders join to take advantage of favorable market conditions. With an ever-increasing inflow of traders, the exchange can execute trades faster, more cheaply, and in turn, even greater numbers become incentivized to join. These "network effects" underlie the core enterprise of an exchange.[68] They promote

[58] Mahoney, *supra* note [57], 1459-1462; Seligman, *supra* note [57], 73-76.

[59] Mahoney, *supra* note [57], 1459-1462. On the benefits of standardization, Craig Pirrong, *A Theory of Financial Exchange Organization*, 43 J. L. & ECON. 437, 439–40 (2000).

[60] Mahoney, *supra* note [57], 1461-1462 (detailing rules about the creditworthiness of members).

[61] Mahoney, *supra* note [57], 1461.

[62] From the 1840s, so-called Boards of Trade – with the Chicago Board of Trade being the most prominent – created agreed-upon systems for exchanging commodities, including mechanisms for grading, and inspecting goods. See for example, Joseph Santos, *A History of Futures Trading in the United States, EH.Net*, https://eh.net/encyclopedia/a-history-of-futures-trading-in-the-united-states/.

[63] Mahoney, *supra* note [57], 1461.

[64] Mahoney, *supra* note [57], 1461.

[65] Mahoney, *supra* note [57], 1457-1458.

[66] Dan Awrey, *The Mechanisms of Derivatives Market Efficiency*, 91 N.Y.U. L. REV. 1104, 1107-1108 (2016) On the risks of bilateral trading and its inefficiencies); Pirrong, *supra* note [59] 437, 439–40 (detailing how bilateral trading results in transaction costs).

[67] Bengt Holmström & Jean Tirole, *Market Liquidity and Performance Monitoring*, 101 J. POL. ECON. 678 (1993) (noting the monitoring role facilitated by liquid stock markets, with prices able to offer surveillance).

[68] *See e.g.,* Marco Pagano, *Trading Volume & Asset Liquidity*, 104 Q.J. ECON. 579 (1995) (highlighting network externalities for exchanges in the context of exchange design); Ananth Madhavan, *Market Microstructure: A Survey*, 3 J. FIN. MKTS. 205, 216–19 (2000) (noting the importance of network externalities for exchanges); Nicholas Economides, *Network Economics with Application to Finance*, 2 FIN. MARKETS, INST. & INSTR. (1993), 89-91(highlighting network externalities for promoting certainty); See also, Robert Ahdieh, *Making Markets: Network Effects and the Role of Law in the Creation of Strong Securities Markets*, 104 S. CAL. L. REV. 277, 285-288 (2003).

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efficiencies as trading becomes cost-effective, faster, and driven by volumes of traders that introduce varied perspectives, insights, and information.[69] Powerful network effects help generate "liquid" marketplaces, where transactions flow smoothly and inexpensively, prices are stable, and primed to convey information about the present value of assets like stocks or commodities.[70]

This need for network effects explains why exchanges have, from their earliest years, sought to create order, oversight, and discipline within their venue.[71] For a start, entry criteria for traders and goods can assure that counterparties on the venue have financial means, legal capacity, and good standing, and that what they are trading is economically and legally sound. Each participant must feel confident that their bargain will be enforced, mitigating the need to discount capital for the risks of misconduct or breach.[72] Because an exchange verifies the quality of claims traded, traders save themselves the time and money involved in performing this task.[73] This leaves them well-situated to spend on research that adds to the quality of prices produced on the venue, rather than duplicative investigation into the basic attributes of the claim.[74]

Further, mandatory rules and their enforcement promotes efficiency within the trading process. Traders know how to submit offers to buy and sell, when these offers become binding, the penalties for breach, and how execution will take place (i.e., to transfer assets from the seller to the buyer in return for cash). Standard-form contracts can help make claims more tradable. For commodities, an exchange's contract might specify the quantities of the good being traded (e.g., ten barrels of crude oil), its quality (e.g., grade and purity levels), when delivery must be made (e.g., on any day during the specified delivery month), and how it should be stored (e.g., in designated warehouses). These rules provide predictability, clarity, and efficiency. They save traders from having to negotiate terms (e.g., quality and quantity of goods being traded,

[69] Jonathan R. Macey & Hideki Kanda, *The Stock Exchange as a Firm: The Emergence of Close Substitutes for the New York and Tokyo Stock Exchanges*, 75 CORNELL L. REV. 1007, 1012-1014 (observing that liquid and efficient markets are interrelated); See also, Ronald Gilson & Reinier Kraakman, *The Mechanisms of Market Efficiency*, 70 VA. L. REV. 549 (1984); Awrey, *supra* note [66].

[70] Eugene Fama, *Efficient Capital Markets: A Review of Theory and Empirical Work*, 25 J. FIN. 383, 383 (1970) ("A market in which prices always 'fully reflect' available information is called 'efficient.'"). On critiques of efficient markets, see, e.g., ANDREI SCHLEIFER, INEFFICIENT MARKETS: AN INTRODUCTION TO BEHAVIORAL FINANCE (2000). In the legal literature, Ronald Gilson and Reinier Kraakman have detailed dynamics that help make public markets more informationally efficient. See generally, Gilson & Kraakman, *supra* note [69]. On discussions of liquidity see for example, Douglas J. Elliott, *Market Liquidity: A Primer*, ECON. STUD. BROOKINGS, 1, 3–5 (2015).

[71] Mahoney, *supra* note [57], 1454, 1457.

[72] On discounting, see generally Aswath Damodaran, *Equity Risk Premiums: Determinants, Estimation, and Implications*, 11–14 (2013).

[73] For discussion, Yesha Yadav, *Oversight Failure in Securities Markets*, 104 CORNELL L. Rev. 101, 113-115 (2019), 1826-1827.

[74] Adam C. Pritchard, *Markets as Monitors: A Proposal to Replace Class Actions with Exchanges as Securities Fraud Enforcers*, 85 VA. L. REV. 925 (1999) (arguing that strong exchange oversight has positive benefits for attracting top companies to list and smart traders to the venue).

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how they need to be stored). These contracts become easier to price owing to standardization and a broad understanding of terms.[75]

Finally, exchange discipline can offer assurance that bargains between traders are executed exactly as agreed. Contracts decisions are thus honored even under difficult circumstances. If an airline has contracted to purchase 100 barrels of oil at $100, it gets the full benefit of this contract even if the price of oil skyrockets to $200 per barrel. In such circumstances, a seller might decide to breach the agreement to sell its oil elsewhere. But such an outcome would spell doom for the exchange if contracts could be breached whenever parties find a more lucrative alternative. Modern exchanges generally utilize specialist institutions called clearinghouses to ensure that contracts are settled as agreed to between the parties. Once a contract matches on an exchange and binds the parties, it moves to a clearinghouse. The clearinghouse then takes over and guarantees to each buyer and seller that the deal will be executed. If a party fails and defaults, the clearinghouse has in-built protections and still follows through on the bargain, procuring either goods or cash, and preventing the consequences of the default from causing distress to those whose bargains risk collapse.[76]

As Chris Brummer argues, in establishing a particular regulatory environment within their venues – designed to control entry, maintain trading and secure finality for trades – exchanges have become a kind of marketplace for selling "law" to market participants.[77] Its distinct oversight environment represents a specific type of product that singles that exchange out as worthwhile to investors and public regulators. To be sure, this does not mean that an exchange necessarily has to offer the strictest environment to stand out as attractive. As Brummer notes, investors and companies will seek out venues for their ease of trading (liquidity) as well as for preferences they might have for the broader national legal environment linked to a particular venue.[78] Nevertheless, a rulebook and a purpose-built regulatory ecosystem has come to constitute a critical component of exchange design in general – as well as of an exchange's business model in particular. This interconnection between an exchange and its regulatory environment more closely tethers its ability to offer a bustling marketplace to its capacity to also deliver a model of oversight.

B. How Exchanges Regulate Financial Markets

[75] *See e.g.*, Gus de Franco et al., *Similarity in Bond Covenants*, Working Paper (2016) (detailing the benefits of standardization for corporate bond liquidity).

[76] *See generally*, Awrey & Macey, *infra* note [312].

[77] Brummer, *supra* note [52], 1437-1438.

[78] Brummer, *supra* note [52], 1437-1439.

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Modern financial exchanges require those using them to agree to a detailed, prescriptive rulebook that, in theory, affords the exchange close control over the everyday functions of a trader, listed company, or asset.[79]

These rulebooks typically force market participants to undergo thoroughgoing changes to their governance and operational workings. To take one example, those seeking membership to trade on the NYSE must establish internal compliance mechanisms to monitor adherence to applicable securities regulations and exchange rules.[80] Members must create offices that check that the firm is acting in conformity with laws, regulations, and exchange rules. They must designate qualified personnel within these offices who are accountable for maintaining and supervising the member's compliance.[81] This means keeping track of written communications (e.g., emails) concerning the member's investment business. Where members represent clients – and trade on behalf of these clients on the exchange – they must keep a detailed set of processes and procedures to safeguard their client relationships (e.g., ensure proper communication with clients), any client funds that are being held (e.g., segregate client money from a member firm's money), and deal with possible fallout (e.g., have systems to handle client complaints).[82]

Exchanges are also prescriptive about the design and quality of trading systems that connect a trading member to the venue. These systems perform logistical tasks that create risks for others should they fail to work. Traders are thus subject to granular constraints on how they behave and on the content and quality of their trading infrastructure. To take one example, the Chicago Mercantile Exchange (CME) prescribes how traders use its systems.[83] It prohibits submitting bids and offers on the exchange that are liable to confuse other traders.[84] The exchange scrutinizes and punishes a range of disruptive and dangerous practices.[85] Orders must be submitted with a real intent to execute, not simply to spoof the system.[86] Traders are forbidden from entering into deals where the purpose lies in taking no economic risk whatsoever – a "wash" where the same person may be on both sides of a buy and sell transaction.[87] Importantly, those seeking to transact on the CME must make sure they take care to use systems that do not flood the exchange with messages, intentionally or otherwise, causing harm to other users, the smooth running of the venue, and

[79] NYSE RULES, RULES - ALL NYSE GROUP EXCHANGES, https://www.nyse.com/regulation/rules.

[80] NYSE RULES, ANNUAL CERTIFICATION OF COMPLIANCE AND SUPERVISORY PROCESSES, RULE 3130.

[81] NYSE RULES, SUPERVISION, RULE 3110; NYSE RULES, SUPERVISORY CONTROL SYSTEM, RULE 3120.

[82] *Id.*

[83] CME GROUP, USER HELP SYSTEM: ENTER ORDERS, https://www.cmegroup.com/tools-information/webhelp/cmeone-cmed-mobile/Content/EnterOrders.html.

[84] CME GROUP, RULEBOOK: TRADING INFRACTIONS, RULE 514A.2.

[85] The rulebook covers a range of stipulations across conduct, governance and infrastructure. See in general, CME GROUP, RULEBOOK. https://www.cmegroup.com/rulebook/CME/.

[86] CME GROUP, RULEBOOK: DISRUPTIVE PRACTICES PROHIBITED, RULE 575.

[87] CME GROUP, RULEBOOK: WASH TRADES PROHIBITED, RULE 534.

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its technology.[88] Access to the CME's communication engines is controlled. The CME vets every order that it receives to make sure that this message's internal data and logic are in conformity with those of the exchange.[89] Finally, it retains expansive authority to respond in cases where trading ends up disrupted – for example, by instituting temporary halts to trading.[90]

Among the oversight responsibilities performed by exchanges, perhaps the most visible is the vetting they provide to new companies and assets that list for trading on their venue. As noted earlier, this quality control saves investors the cost of having to investigate using their own private resources.[91] Initially designed in the 1860s to check the basic firm characteristics of an issuer, corporate listing requirements have gradually grown more searching to create a framework of conditions that governs minute aspects of a company's life.[92] Importantly, this intensification has meant that the NYSE and Nasdaq – the two major U.S. listing exchanges for public equity – have come to play a much more influential role as supervisors in the everyday lives of listed companies.[93] Following infamous corporate governance scandals in the early 2000s (e.g., Enron, WorldCom), listing rules have become powerful for instituting fundamental changes to C-suite practices.[94] For example, the board of directors of a public company must comprise a majority of independent directors. In the case of companies listed on the NYSE, the audit, nominating and compensation committees must be entirely comprised of independent directors.[95]

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[88] CME GROUP, RULEBOOK: DISRUPTIVE PRACTICES PROHIBITED, RULE 575.

[89] CME Group, *What Happens When You Submit an Order*, https://www.cmegroup.com/education/courses/things-to-know-before-trading-cme-futures/what-happens-when-you-submit-an-order.html

[90] CME GROUP, RULEBOOK: GLOBEX COMMAND CENTER, RULE 579 (giving the CME's Global Command Center authority to take a variety of steps, including trading halts, in order to restore order and maintain the integrity of trading and prices).

[91] *See generally*, Roberta S. Karmel, *The Future of Corporate Governance Listing Requirements*, 54 S.M.U. L. REV. 325, 326-333 (2001) (detailing listing standards as a quasi-substitute for federal regulation and noting greater intensity of listing conditions on the NYSE from 1869 onward). On international competition and stock exchange listing choices, Brummer, *supra* note [52], 1435-1436; Scholars have debated the role and impact of national regulatory systems on the flows global investment. For example, see generally, Jack Coffee, *Racing Towards the Top?: The Impact of Cross-Listings and Stock Market Competition on International Corporate Governance*, 102 COLUM. L. REV. 1757 (2002) (noting the positive impact of stricter listing and disclosure requirements for attracting capital); Stephen Choi and Andrew Guzman, *National Laws, International Money: Regulation in a Global Capital Market*, 65 FORDHAM L. REV. 1867 (1997) (detailing how regulatory change within single jurisdictions can impact cross-border investment).

[92] Karmel, *supra* note [91], 326-328. For a skeptical account of the impact of listing requirements, see, Macey & Kanda, *supra* note [69], 1039-1040 (noting that the SEC displaced the NYSE as a source for corporate governance standards].

[93] The American Stock Exchange was a major listing exchange capturing smaller listings and one that did not qualify for the NYSE. For discussion, Onnig Dombalagian, *Demythologizing the Stock Exchange; Reconciling Self-Regulation and the National Market System*, 39 U. RICH. L. REV. 1069, 1078–79 (2005).

[94] This was triggered by the passage of the Sarbanes-Oxley Act, Sarbanes-Oxley Act, PUB. L. 107-377 (2002) in response to the scandals of the early 2000s.

[95] For discussion of differences between the Nasdaq & NYSE, Mark Kuehn, *Sarbanes-Oxley Act and New NYSE and NASDAQ Listing Requirements: Consequences for Private Equity and Mergers and Acquisitions Transactions*, Gibbons Law (Jan. 7, 2004), https://www.gibbonslaw.com/resources/publications/sarbanes-oxley-act-and-new-nyse-and-nasdaq-listing-requirements-consequences-for-private-equity-and-mergers-and-acquisitions-transactions-01-07-2004.

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As Robert Thompson observes, this role has brought the NYSE and Nasdaq into a collaborative supervisory relationship with federal regulators, notably the SEC. This cooperation extends to rulemaking as much as it does to enforcement. Exchanges provide the real-world problem-solving entailed in bringing a policy goal to fruition – taking a cue from the SEC's agenda to create a listing rule that can be applied across the largest companies.[96] This can be helpful to policymakers where exchanges have flexibility to craft rules that a public regulator might struggle to pass.[97] For example, in August 2021, Nasdaq received approval from the SEC to implement a rule to promote greater board diversity. The rule requires listed companies to disclose how diverse the company's board is and, if needed, to explain why the board does not include at least two members from underrepresented communities.[98] As Chris Brummer and Leo Strine posit, this rule held the potential to force a swath of companies to reckon with limited diversity in leadership.[99] But its emergence has also drawn into relief earlier failures on the part of public regulators to encourage diversity and inclusion within public companies. Nasdaq's effort follows the SEC's own attempts to encourage board diversity through disclosure requirements that commentators criticized as vague, nebulous, and ultimately ineffectual.[100] With this presence in boardrooms, exchange listing rules have created a national overlay to otherwise state-dominated corporate laws.[101]

Crucially, exchanges deploy a range of hard and soft coercive methods to enforce their rules. In exercising formal power, exchanges stipulate a range of penalties to secure compliance. These operate along a scale of seriousness and reflect an exercise of discretion and judgment on the part of the exchange. For example, the NYSE sets out a list of possible punishments when a member or supervised entity, is in breach of the rulebook. At its lighter end, the exchange might privately censure the user, or impose a fine. In matters of increasing seriousness, those in breach risk suspension or expulsion and a potential bar on their dealings with other members. The NYSE can also impose any sanction that it considers appropriate.[102] For public companies, the ultimate censure means being delisted and losing their ability to raise capital in public markets.[103]

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[96] Robert Thompson, *Collaborative Corporate Governance: Listing Standards, State Law and Federal Regulation*, Working Paper (2003), 11-12.

[97] Thompson, supra note [96], 6-7.

[98] NASDAQ, CORPORATE GOVERNANCE REQUIREMENTS, RULES 5605(F) & 5606; U.S. SEC. & EXCH. COMM'N, STATEMENT ON THE COMMISSION'S APPROVAL OF NASDAQ'S PROPOSAL FOR DISCLOSURE ABOUT BOARD DIVERSITY AND PROPOSAL FOR BOARD RECRUITING SERVICE, https://www.sec.gov/news/public-statement/gensler-statement-nasdaq-proposal-disclosure-board-diversity-080621 (Aug. 6, 2021) (diverse generally refers to a person who self-identifies as a part of one or more of the following communities: female, underrepresented minority, LGBTQIA+). Brian Breheny et al., *SEC Approves Nasdaq Board Diversity Listing Standards*, Skadden Insights, Sept. 28, 2021; Chris Brummer & Leo Strine, *Duty and Diversity*, 75 VAND. L. REV.1, 52-60 (2022) (detailing various SEC, state-law, and private initiatives to bring diversity to boards).

[99] Brummer & Strine, *supra* note [98], 59-60.

[100] Brummer & Strine, *supra* note [98], 53-54.

[101] Karmel, *supra* note [91], 325-326; Thompson, *supra* note [96], 5-7.

[102] NYSE, SANCTIONS, RULE 8310.

[103] NYSE REGULATION, DELISTINGS, https://www.nyse.com/regulation/delistings.

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Beyond the hard power of formal sanction, exchanges are adept at harnessing the soft power of shame. Sanctions are publicized.[104] With the exception of minor offenses, the NYSE retains discretion to put information about punishment on its website and to furnish details about infractions to the public.[105] As David Skeel has written, shaming can be highly effective as a form of discipline, particularly within smaller communities where the value of reputation carries special currency.[106]

Those disciplined by an exchange can suffer severe economic consequences. Studies have examined the real-world impact of de-listings – and point to a litany of grave outcomes for companies ousted from the exchange. According to one looking at de-listings from the NYSE between 1995-2005, the share price of affected companies crashed by 50%.[107] Another study on Nasdaq de-listings similarly found that impacted companies saw share prices lose half their value, and the transaction costs associated with buying and selling shares tripled.[108] Even when companies are not de-listed but just subject to a notice of a possible (and often fixable) delinquency – negative consequences follow swiftly. One study, examining such Nasdaq notices showed a disapproving response from the marketplace on the disclosure of such a notice, underscoring the reputational harm accruing to those finding themselves under the cloud of a potential (not always actual) breach of exchange rules.[109]

C. The Public-Private Model in Financial Markets Oversight

Public regulation thus expressly harnesses exchange oversight as a part of the larger supervisory framework for financial markets.[110] Present-day regulation represents a longstanding public-private bargain, where exchanges are afforded considerable power, judgment, and authority, while falling under a dedicated system of public oversight.[111]

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[104] *See for example,* NYSE REGULATION, DELISTINGS, https://www.nyse.com/regulation/delistings.

[105] NYSE, RELEASE OF DISCIPLINARY COMPLAINTS, DECISIONS AND OTHER INFORMATION, RULE 8313.

[106] See, for example, David Skeel, *Shaming in Corporate Law*, 149 U. PENN. L. REV. 1811, 1811-1812 (2001).

[107] Jonathan Macey, Maureen O'Hara & David Pompilio, *Down and Out in the Stock Market: The Law and Economics of the Delisting Process*, 51 J.L. & ECON 683 (2008). Showing a negative effect on share price after a company announces delisting, see, Gary Sanger & James Peterson, *An Empirical Analysis of Common Stock Delistings*, 25 J. FIN. & QUANTITATIVE ANALYSIS 261 (1990).

[108] Venkatesh Panchapagesan & Ingrid Werner, *From Pink Slips to Pink Sheets: Market Quality Around Delisting from Nasdaq* (EFA 2004 Maastricht Meetings, Working Paper No. 4572, 2004).

[109] Carol Frost, Joshua Racca & Mary Stanford, *Shareholder Wealth Effects of Corporate Governance Deficiencies on Nasdaq* 3 (Working Paper, 2017). For discussion, Yadav, *supra* note [73], 1821-1823.

[110] Mahoney, *supra* note [57], 1457-1560; Thompson, *supra* note [96], 9-10; Roberta S. Karmel, *Should Securities Industry Self-Regulatory Organizations Be Considered Government Agencies*, 14 STAN. J.L. BUS. & FIN. 151, 153-160(2008).

[111] Dombalagian, *supra* note [93], 1079-1082.

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Classified as "self-regulatory organizations" (SROs), exchanges act as quasi-regulators with surveillance and disciplinary authority, but themselves subject to checks and balances designed to curb tendencies towards misconduct, risk-taking, manipulation and self-dealing within their venues.[112] As such, while exchanges have considerable authority to make rules for their markets, their proposals for new rulemaking must first be published and approved by the SEC.[113] Venues administer a range of disciplinary sanctions. But, they may only do so in accordance with standards of due process.[114] Exchanges enjoy a wide berth when organizing their own operations and governance.[115] However, regulation demands that access to exchange membership and control be non-discriminatory and fair.[116] Crucially, exchanges enjoy qualified immunity from lawsuits owing to their regulatory status.[117] Still, this protection does not extend to actions arising out of their commercial activities.[118] Ultimately regulators can punish misbehaving exchanges by rescinding their authority to operate.[119]

Even as exchanges enjoy this perch of responsibility, policy recognizes that they represent imperfect monitors.[120] In other words, exchange oversight comes with problems. Scholars point to shortcomings that can skew the effectiveness of discipline. Most systematic is the fact that financial exchanges are mostly run as for-profit institutions, whose ownership rests in the hands of dispersed shareholders.[121] Their own bottom line thus relies, in part, on the business generated by traders or companies that list their securities, paying fees and attracting the liquidity that builds an exchange's brand and popularity.[122] It

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[112] This framework was set out under the Securities Exchange Act 1934, and through Section 4 of the Act, which accorded the Securities and Exchange Commission the power to implement the Act and its provisions, including with respect to self-regulatory organizations. 15 U.S.C. § 78d (2000); For a history and analysis, Dombalagian, *supra* note [93], 1073-1082.

[113] Securities Exchange Act of 1934, ch. 404, § 6(c), 48 Stat. 881, 886 (1934); 15 U.S.C. § 78w(a)(3)(2000); For discussion, Dombalagian, *supra* note [93], 1076-1077.

[114] Dombalagian, *supra* note [93], 1081; Exchange Act § 6(a), 15 U.S.C. § 78f(b) (2000).

[115] Securities Exchange Act of 1934, ch. 404, § 6(c), 48 Stat. 881, 886 (1934); For discussion, Dombalagian, *supra* note [93], 1076-1077.

[116] Dombalagian, *supra* note [93], 1081.

[117] Sparta Surgical Corp. v. NASD, Inc., 159 F.3d 1209, 1213 (9th Cir. 1998) (noting that exchanges have immunity for acts undertaken as part of their regulatory functions); Barbara v. New York Stock Exchange, Inc., 99 F.3d 49 (2d Cir. 1996) (highlighting immunity for exchanges discharging their disciplinary power).

[118] Weissman v. NASD, Inc. (Weissman IV), 500 F.3d 1293, 1299 (11th Cir. 2007); City of Providence v. BATS Global Markets, Inc., 878 F.3d 36 (2d Cir. 2017).

[119] 15 USC § 78s(h)(1).

[120] See e.g., Marcel Kahan, *Some Problems with Stock Exchange Based Securities Regulation*, 83 VA. L. REV. 1509 (1997) (noting conflicts of interest in exchanges operation); Karmel, *supra* note [91] (positing whether competition between exchanges might drive down the quality of exchange oversight); Macey & O'Hara, *supra* note [49]; SECURITIES & EXCH. COMM., CONCEPT RELEASE CONCERNING SELF-REGULATION, 69 Fed. Reg. 71256 (proposed Dec. 8, 2004), available at http://sec.gov/rules/concept/34-50700.pdf (observing that self-regulation is a key part of US regulation, while posing risks of its own as a model for oversight).

[121] *See e.g.*, INTERCONTINENTAL EXCHANGE, ANNUAL REPORT (2021), 4-13.

[122] Stanislav Dolgopolov, *The Maker-Taker Pricing Model and its Impact on the Securities Market Structure*, 8 VA. L. BUS. REV. 231 (2014) (highlighting the practice of exchanges paying traders to provide liquidity); Hendrik Bessembinder, Jia Hao & Michael Lemmon, *Why Designate Market Makers? Affirmative Obligations and Market Quality* (Working Paper, 2011) (showing the significance of liquidity providers for exchanges).

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follows that exchanges confront tricky trade-offs when exercising discipline. Investing in oversight pays off when it generates liquidity, reduced scrutiny from a regulator, as well as positive network effects where reputational gains attract business.[123] But it can be loss-making. Expelling traders, or aggressively delisting companies can make an exchange unpopular, costing business.[124] This can result in cheaper, make-do surveillance, and adjudication mechanisms becoming attractive in place of stringent design choices that eat into margins.[125]

These conflicts have deepened in recent years, with exchanges becoming highly commercial firms offering a range of (often pricey) products to those using their venue.[126] Beyond simply providing platforms on which parties transact or list their securities, exchanges have developed lucrative product offerings to accompany their core business.[127] To take one example, exchanges routinely sell proprietary data feeds.[128] As Jonathan Brogaard, James Brugler and Dominik Roesch note, exchanges are ideally placed to extract maximum rents from such data offerings owing to the monopoly they exercise over the data produced by their own venue.[129] This has led to data products becoming significant sources of revenue, with Nasdaq enjoying 20% of its revenue from this business line.[130] According to a survey conducted by the Securities and Financial Markets Association (SIFMA), data prices have, by some measures, increased by as much as 2,916% between 2010-2018 even as the costs of trading have decreased.[131]

To mitigate these concerns, exchanges have attempted various institutional fixes. A number of exchanges rely, in part, on the Financial Industry Regulatory Authority (FINRA), an industry overseer of broker-dealer securities firms, to discharge some of their supervisory functions.[132] The NYSE operates its supervisory arm, NYSE Regulation, as a non-profit to carry out its oversight responsibilities.[133] Despite these efforts, however, the core problem of exchanges being economically dependent on their users remains pervasive.

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[123] Brummer, *supra* note [52].

[124] For discussion, Yadav, *supra* note [73], 1843-1845.

[125] Andreas M. Fleckner, *Stock Exchanges at the Crossroads*, 74 FORDHAM L. REV. 2541, 2546-2547(2006) (detailing the shift toward profit-orientated approaches within exchange groups).

[126] On exchange competition and the impact on market quality, see generally, Yadav, *Oversight Failure*, *supra* note [73].

[127] See generally, e.g., Yesha Yadav, *Insider Trading and Market Structure*, 63 UCLA L. REV. 968 (2016) (describing the practice of exchanges offering expensive data feeds and colocation space next to exchange servers).

[128] Yadav, *Insider Trading*, *supra* note [127], 998-999.

[129] Jonathan Brogaard, James Brugler & Dominik Roesch, *Competition and Exchange Data Fees*, Working Paper (2022), 1.

[130] Brogaard, Brugler & Roesch., *Competition*, *supra* note [129], 1.

[131] SIFMA, *The Cost of Investing Is Going Down, So Why Are Market Data Fees Rising?*, Jan 31, 2019, https://www.sifma.org/resources/news/real-clear-markets-the-cost-of-investing-is-going-down-so-why-are-market-data-fees-rising/.

[132] John McCrank, *Wall Street Watchdog FINRA to Monitor BATS' Markets*, REUTERS, Feb. 6, 2015; NASDAQ, RULEBOOK: NASDAQ REGULATORY CONTRACT WITH FINRA (RULE 1001), https://listingcenter.nasdaq.com/rulebook/nasdaq/rules/Nasdaq%20General%203.

[133] NYSE, NYSE REGULATION, https://www.nyse.com/regulation.

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It casts doubt on the motivation of exchanges to exercise discipline optimally in a way that best matches the risks posed, where decisions about surveillance and censure also create costs for profitability within a competitive ecosystem for trading services.[134]

The consequences of these conflicts have emerged in actions to punish exchanges for lax or sub-par supervision. In 2005, for example, the NYSE was disciplined for a years-long failure to sanction abusive practices by its most influential traders, who were opportunistically using their special access to data to transact ahead of clients.[135] In 2014, the SEC fined NYSE for failing to implement or enforce a number of rules that would disadvantage prominent traders.[136] In one infraction, the NYSE concluded differential terms between itself and traders to locate their servers next to those of the exchange.[137] Despite the statutory mandate that an exchange behave equitably, the NYSE permitted trading information to be disseminated to its floor traders ahead of others.[138] This allowed floor traders to gain over an hour's head-start in discovering how the market was moving and getting to craft their strategies in response.[139]

These handful of examples draw into the relief the key takeaways of this Part. They highlight the significance of exchanges as central to the regulatory architecture in modern financial markets – and the complex position they occupy as overseers of a market in which they are also founding participants. Exchanges, in other words, represent repositories of expertise, experience, power, and credibility built from their position as hubs of economic activity, bringing traders, companies and assets together, and using network effects to promulgate rules and industry norms. It follows that regulators have long designated them as SROs, afforded certain privileges in return for the delivery of oversight as public good. Although vulnerable to conflicts, where fallout can result in economic damage, exchange discipline has nevertheless exerted a transformative impact on financial market design and regulation.

III. EXTENDING EXCHANGE OVERSIGHT TO CRYPTO-MARKETS

This Part makes the case for mandating private self-regulation by exchanges in cryptocurrency markets. Cryptocurrencies have, in short order,

[134] See generally, Yadav, *Oversight Failure*, *supra* note [73**Error! Bookmark not defined.**]

[135] Securities and Exchange Commission, NYSE Agrees to Settlement With SEC, Including Censure, Cease and Desist Order, $20 Million Fund for Regulatory Auditor, and Audio-Video Surveillance, Press Release, Apr. 12, 2005, https://www.sec.gov/news/press/2005-53.htm.

[136] Securities and Exchange Commission, SEC Charges NYSE, NYSE ARCA, and NYSE MKT for Repeated Failures to Operate in Accordance with Exchange Rules, Press Release, May 1. 2014, https://www.sec.gov/news/press-release/2014-87.

[137] Jeff Cox, *New York Stock Exchange to Pay $4.5M to Settle SEC Charges*, CNBC, May 1, 2014 (noting that, though the financial penalty was light, the reputational cost to the NYSE was high).

[138] Cox, *supra* note [137].

[139] Cox, *supra* note [137].

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challenged as well as co-opted the traditional financial system as part of a lively wave of digital and financial innovation over the last decade.[140] Since Satoshi Nakamoto's 2008 white paper introducing Bitcoin and its grounding concepts, cryptocurrencies have made dramatic gains in financial markets – as well as, significantly, within the public's imagination.[141] Much attention has been popularly directed toward Bitcoin, Ether, as well as splashy altcoins like Dogecoin.[142] But animating ideas behind Bitcoin have also given rise to a sophisticated array of products seeking to replicate traditional forms of lending, investment, and trading using more decentralized infrastructure – so-called decentralized finance (defi).[143] Rather than relying on usual intermediaries like banks, defi seeks to enable users to access financial services on a peer-to-peer basis aided by automated code – or smart contracts – to facilitate transactions.[144]

In situating cryptocurrency exchanges within this ecosystem, this Part makes the following points. First, despite the emphasis on decentralization, cryptocurrency exchanges constitute highly centralizing and centralized firms.[145] As points of convergence for traders, assets, defi applications and blockchains, and showcasing a large economic footprint, exchanges occupy a unique position from which to exercise monitoring and discipline.[146] Like traditional exchanges, they possess special advantages with which they can perform oversight relatively cheaply and effectively. Secondly, similar to financial exchanges, cryptocurrency trading platforms can gain from private discipline. As centralized firms, fallout from a risky and shoddily governed marketplace reverberates deeply and can concentrate risk within their

[140] *See e.g.*, Kevin Roose, *The Latecomer's Guide to Crypto*, N.Y. TIMES, https://www.nytimes.com/interactive/2022/03/18/technology/cryptocurrency-crypto-guide.html?action=click&module=RelatedLinks&pgtype=Article (noting that 20% of all Americans and 36% of millennials own crypto); Silla Brush, *BlackRock Teams Up With Coinbase in Crypto Market Expansion*, BLOOMBERG, AUG. 4, 2022.

[141] Satoshi Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, https://bitcoin.org/bitcoin.pdf; Roose, *supra* note [140]; Arijit Sarkar, *Satoshi Nakamoto's Bitcoin White Paper is Now a 13-year-old Teenager*, COINTELEGRAPH, Oct. 31, 2021. On the impact within mainstream finance, *see e.g.,* Tara Siegel Bernard, *Everyone Has Crypto FOMO, but Does It Belong in Your Portfolio?* N.Y. TIMES, Mar. 22, 2022; Brush, *supra* note [140].

[142] *See e.g.*, Chris Katje, *Internet Rumor Connects Vitalik Buterin and Elon Musk to Creating Shiba Inu: Here's How*, BENZINGA, Apr. 22, 2022.

[143] On defi and its applications, Brummer, *Disclosure, Dapps and DeFi*, supra note [20], 2-5; COINGECKO, TOP 100 DEFI COINS BY MARKET CAPITALIZATION, https://www.coingecko.com/en/categories/decentralized-finance-defi. David Yaffe-Bellany, *Celsius Network Leads Crypto Markets into Another Free Fall*, N.Y. TIMES, Jun. 13, 2022 (detailing the explosive growth of defi applications – here discussing Celsius, a crypto-savings firm and its collapse). On smart contracts, their operation and risks, Kevin Werbach and Nicholas Cornell, *Contracts Ex Machina*, 67 DUKE L. J. 313 (2017).

[144] Brummer, *Disclosure, Dapps and DeFi*, supra note supra note [20], 2-4; On centralization in defi, Douglas Cumming et al., *Decentralized Finance, Crypto Funds, and Value Creation in Tokenized Firms*, Working Paper (May 2022) (noting the centralizing impact of crypto funds as intermediaries in defi).

[145] For discussion, see e.g., Kristin Johnson, *Decentralized Finance: Regulating Cryptocurrency Exchanges*, 62 WM. & MARY L. REV. 1911, 1953-1954 (2021) (detailing the centralized nature of cryptocurrency exchanges).

[146] See e.g., Ben Bartenstein, *Binance Limits Russia Services After EU Sanctions on Crypto,* BLOOMBERG, Apr. 10, 2022 (on efforts by regulators to harness crypto-exchanges within the sanctions framework).

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institution.[147] More cautiously, as with traditional exchanges, there are reasons to be realistic and circumspect in relying entirely on cryptocurrency platforms for oversight. Coming of age with relatively low barriers to entry, the market for trading services in cryptocurrencies is fiercely competitive. This has eroded exchange incentives to cooperate with one another and to favor private profit-seeking above supervision as overriding interest. Further, a deeply centralized business model – conjoining multiple trading functions alongside commercial interests – has given rise to conflicts of interests that appear embedded within firm structures. Rather than expecting cryptocurrency exchanges to engage in oversight of their own accord, this Part demonstrates the need for a legal mandate for self-regulation as a way to foster incentives that better align private interests of venues with the public good.

A. Exchanges and Cryptocurrency Market Structure

As detailed in this Part, cryptocurrency exchanges have shown themselves as essential to the creation of a market for digital assets. Even more so than traditional venues, crypto trading platforms provide specific advantages for users by bypassing the technical and operational challenges involved in transacting on blockchains. Exchanges attract users by offering a more user-friendly, predictable, and fast trading environment that can be detached from blockchains, and capable of offering users certainty of execution and clarity in costs, alongside a suite of lucrative side-services.

1. Some Basics of Crypto-Market Structure

First, some basics. Bitcoin seeks to transfer value directly between two users anywhere in the world outside of traditional intermediation. To do this, in the absence of a conventional central player, Bitcoin's own operating system must generate safety, integrity, privacy, and efficiency computationally.[148]

In a typical payment, a bank will generally verify a customer's account details, check their identity, ensure that the sender's account details are valid, and stand behind a transaction. Within Bitcoin's system, however, these tasks must be approximated using computer code. To do this, the Bitcoin protocol relies on four key concepts. First, information is encrypted.[149] Instead of being known by their real-world names and personal details, users are recognized by

[147] *See e.g.*, Jordan Valinsky, *Coinbase Lays Off 18% of its Workforce and Warns of 'Crypto Winter,'* CNN, Jun. 14, 2022.

[148] Nakamoto, *supra* note [141], 1-2.

[149] Nakamoto, *supra* note [141], 2, 6.

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encrypted, computer-generated pseudonymous addresses.[150] Messages sent to a user's public address can only be unlocked using a unique, encrypted private password (or key). This strict correspondence between the public address and its private key secures the user's place within the network and offers assurance that the user is entitled to the value in their account.[151]

Secondly, verification of transaction information is undertaken by consensus among participating users within the Bitcoin network.[152] Rather than a bank being responsible for checking payment details, the task falls to the network as a whole. Participating computers – acting as a "node"– check whether each transaction meets pre-set criteria for determining whether the transfer is valid (e.g., has the user signed-off with the private key to match their public address? Do they have sufficient Bitcoin within their account? Is the Bitcoin being transferred legitimate – or does it represent an attempt to maliciously spend the same Coin twice?).[153] Once the transfer gains the threshold of agreement across the distributed network of nodes, it is considered valid and included as part of the network's permanent record (or ledger).[154]

Thirdly, the network relies on transparency and immutability to provide security and certainty. Rather than offer users refunds, discounts, do-overs, or updates on a transfer, as might happen if a bank transfer goes awry, transactions within the Bitcoin network cannot be undone without expenditure of enormous computational power.[155] To provide confidence in its integrity, information about each valid transfer is published to a public record – a ledger – that reflects the state of the network's activities and allows anyone to access data on the flow of value between users. This distributed ledger – colloquially a blockchain – identifies all verified and completed trades, offering public assurance of the network's resilience to manipulation or damage.[156]

Fourthly, the network needs to be the agent of its own protection. In the absence of an intermediary, the network must safeguard its own integrity and continuity. Within a system of globally distributed nodes – that do not know or trust one another – self-preservation represents a tricky challenge. Any node can turn malicious and seek to disrupt the system, for example, by trying to pay itself using coins that have already been spent.

To mitigate this risk, the Bitcoin network creates incentives and pathways for nodes to behave protectively rather than maliciously. It does this

[150] Nakamoto, *supra* note [141], 2, 6.
[151] Nakamoto, *supra* note [141], 2, 6.
[152] Nakamoto, *supra* note [141], 3.
[153] Nakamoto, *supra* note [141], 2-3.
[154] Nakamoto, *supra* note [141], 2-3.
[155] Nakamoto, *supra* note [141], 1-3. Bitcoin works on the concept of probabilistic finality. A transaction becomes more final the more transactions build upon the version of the ledger on which the transaction is featured. As this blockchain becomes longer, and the transaction becomes older, more permanent, and immutable. It would take increasing computing power for a malicious actor to undo a longer series of transactions and create a new version of the ledger. Alexis Gauba, *Finality in Blockchain Consensus*, MEDIUM, Aug. 30, 2008.
[156] Nakamoto, *supra* note [141], 1-3.

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by rewarding nodes that safeguard the network.[157] Nodes that solve a mathematical challenge set by the network win the chance to propose transactions for verification and inclusion on the blockchain. These "miners" are tasked with scouring the pool of incoming orders and proposing to the network only those that look credible and legitimate. Once the network approves these transactions, miners are rewarded for their efforts via freshly minted Bitcoin.[158] In addition, winning miners also get any transaction fees that users sometimes attach to their transfers in a bid to motivate miners to pick their transaction for validation more quickly.[159] Crucially, the economic rewards on offer for those protecting the network helps to stave off competing incentives to disrupt and debase it. Winning new Bitcoin and user fees encourages miners to invest in safeguarding a system that can produce repeat and lucrative profit.[160]

The elegance of this design aside, Bitcoin's utility for real-world, transactions has long been the subject of debate.[161] And its potential for mass adoption has been stymied by the need for more accessible, user-friendly ways to transact in the cryptocurrency. For a start, those seeking out Bitcoin (and comparable cryptocurrencies modeled on its design) face a technological learning-curve. They must set up public addresses on the network, safely store their private keys, access the blockchain, and use the system. Added to this is the challenge of being able to procure/sell coin as needed. Within a universe of pseudonymous players, a person wishing to buy or sell coin has to first seek out another party wishing to trade. This search can be complicated and costly owing to the network's emphasis on privacy. Brokers might be needed. The search can fail, become expensive, or take too long to meet the needs of the moment.

Structural drawbacks also hinder the workings of the blockchain. Users can struggle to see transactions processed at high-speed, efficiently, and predictably. For one, there is no guarantee that transactions will always be picked up by miners from the pool of messages within the network waiting to be verified. Miners have to first dip into this pool, select viable transactions, batch them into blocks and then propose them to the network for validation. A user's transaction might never get picked up – or only get processed after delays. Bitcoin's blockchain has suffered periods of high traffic where large volumes of transactions can end up discarded. When Bitcoin experienced a 15%

[157] Nakamoto, *supra* note [141], 4-5.

[158] This "proof-of-work" system for network protection represents one choice that a blockchain's operating program may make. Another is the "proof-of-stake" method, where those that have the most coins within the system elect to protect the network. Here, a user can "stake" their coins – essentially putting them at risk – and in return, gain a right to assist in validating transactions. Proof-of-stake is viewed as less energy-intensive than proof-of-work, though it also has other drawbacks, like favoring those that are already asset-rich. Ethereum, *Proof-of-Stake*, https://ethereum.org/en/developers/docs/consensus-mechanisms/pos/.

[159] Nakamoto, *supra* note [141], 4-5.

[160] For discussion of transaction fees as necessary aspects of Bitcoin trades, see generally, David Easley & Maureen O'Hara, *From Mining to Markets: The Evolution of Bitcoin Transaction Fees*, 134 J. FIN. ECON. 91 (2019).

[161] *See e.g.*, BANK OF INT'L SETTLEMENTS, ANNUAL ECONOMIC REPORT (2022), 76-77; For discussion, Angela Walch, *Why Bitcoin Fails as Money: An Operational Risk Analysis*, Working Paper (2015) (noting the prevalence of bugs and operational risks that limit Bitcoin's effectiveness as money).

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fall its price in April 2021, system traffic resulted in a pile-up of unprocessed transactions, with almost 130,000 unprocessed transactions accumulating on a single day.[162] Spikes of unprocessed transactions have been visible during the crypto meltdown in Spring 2022, with volumes of unprocessed trades surging around mid-May as crypto-markets experienced crushing losses.[163]

Such uncertainty means that users cannot be sure that their economic goals will be met. For example, transaction costs can increase during busy periods. To increase the chances of uptake, users voluntarily attach fees to their messages in a bid to incentivize miners to pick up their order. As David Easley and Maureen O'Hara observe, transaction fees are now necessary for miners to extract profits. They also show that fees have become virtually essential during periods of high traffic and that without them, transactions will oftentimes fail to attract attention. Users also do not know what kind of fee is likely to be sufficient, requiring them to estimate what others might be bidding.[164] This overall dynamic leads to fragility within the blockchain as users are dissuaded from using the cryptocurrency on account of high costs and a misalignment in results from what they might have been expecting.[165]

In addition to lost and delayed transactions, blockchains themselves are susceptible to breakdown.[166] To be sure, the distributed design of blockchains should be more resilient against the kinds of vulnerabilities created by intermediaries that become single points of failure.[167] However, the fact of being distributed can also result in blockchains experiencing distinct types of operational failure.[168] For example, within a network of public nodes, one or a handful of nodes can become destructive. The 51% attack exemplifies this risk, where a player controlling 51% of the mining power (the "hashrate") within the network can take-over its decision-making and divert value to themselves.[169]

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[162] Tim Hakki, *Bitcoin Transaction Fees Soar to $47 As Market Crashes*, DECRYPT, Apr. 18, 2021, https://decrypt.co/66925/bitcoin-transaction-fees-soar-to-47-as-market-crashes.

[163] Jochen Hoenicke, *Mempool Statistics: Unconfirmed Transaction Count*, https://jochen-hoenicke.de/queue/#BTC%20(default%20mempool),6m,count (for the week around May 16, 2022). On the crypto-meltdown in May 2022, Vildana Hajric, *Bitcoin Fans Seen Bottom-Fishing After Terra-Driven Meltdown*, BLOOMBERG, May. 16, 2022.

[164] See generally, Easley & O'Hara, *supra* note [160].

[165] See generally, Easley & O'Hara, *supra* note [160]. The Ethereum blockchain has faced well-known difficulties during periods of high traffic, that have resulted in a slowdown of the network as well as high transaction (or "gas") fees. For discussion, Marcel Deer, *Will the Ethereum 2.0 Update Reduce High Gas Fees?* COINTELEGRAPH, May 14, 2022.

[166] Felix Irresberger, Kose John, Peter Mueller & Fahad Saleh, *The Public Blockchain Ecosystem: An Empirical Analysis*, Working Paper (2021) (describing different types of blockchain and variety in blockchain design and utility).

[167] See for example, Paolo Saguato, *The Ownership of Clearinghouses: When "Skin in the Game" Is Not Enough, the Remutualization of Clearinghouses*, 34 YALE J. ON. REG. 601, 630–35 (2017) (highlighting the systemic risks posed by clearinghouses).

[168] Nic Carter & Linda Jeng, *DeFi Protocol Risks: The Paradox of DeFi*, Working Paper, 13-17 (Jun. 2021).

[169] Carter & Jeng, *supra* note [168], 16; Eliken Attah, *Five most prolific 51% attacks in crypto: Verge, Ethereum Classic, Bitcoin Gold, Feathercoin, Vertcoin*, CRYPTOSLATE, Apr. 24, 2019, https://cryptoslate.com/prolific-51-attacks-crypto-verge-ethereum-classic-bitcoin-gold-feathercoin-vertcoin/. For the definition of mining power/hashrate, Griffin McShane, *What is a 51% Attack?* COINDESK, https://www.coindesk.com/learn/what-is-a-51-attack/.

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Blockchains can be hacked, corrupting their code. Smaller blockchains are especially vulnerable.[170] Their mining power is often thinner than in larger blockchains (like Bitcoin or Ethereum), making it cheaper for a rogue player to accumulate 51% of the network's hash power.[171] Their programming may not be battle-tested, scalable, and under-programmed to resist a range of risks.[172]

These dangers have materialized in a number of high-profile failures. 51% attacks have disrupted several smaller blockchains, like Firo, Grin, Verge, Ethereum Classic, as well as Bitcoin Gold.[173] As Nic Carter and Linda Jeng write, attacks have often sought to reorganize transactions already validated and recorded on the ledger, altering coin deposits previously credited to users.[174] Attackers might seek to create a new version of the blockchain that reflects a re-allocation of coins to themselves.[175] They may try and spend the same coins twice.[176] In 2021, the Verge blockchain saw almost six months (200 days) worth of transactions vanish from the ledger in a 51% attack, described by observers as the deepest blockchain reorganization in history.[177]

While bigger blockchains like Bitcoin and Ethereum are more resistant to 51% attacks, given superior mining power backstopping their workings, they are still vulnerable to complex hacks and operational failures. Hacks, in particular, have emerged as an ever-present danger. In a famous 2016 example, the Ethereum blockchain experienced a hack directed at an application it hosted, "The DAO." The DAO application had crowdfunded a then-record-breaking $150 million. Weaknesses in the DAO's code, however, were exploited, with 3.6 million Ether – almost 15% outstanding at the time – being stolen. The scale of this theft forced a fundamental reckoning for Ethereum and its norms of immutability and finality. Ethereum's governance responded by instituting a system-wide reset to try and negate the economic effects of the hack, creating a new version of the blockchain that reflected a pre-hack reality. [178]

More recently, a series of high-dollar hacks have targeted so-called bridges – applications that seek to transfer assets between different blockchains like Ethereum and Bitcoin. In 2022, the Ronin and Wormhole bridges were hacked, with attackers making off with a combined loot of almost $1 billion in crypto-assets ($600 million in the Ronin attack; around $320 million in

[170] Carter & Jeng, *supra* note [168], 16.

[171] Carter & Jeng, *supra* note [168].

[172] Irresberger et al., *supra* note [166].

[173] Osato Avan-Nomayo, *Privacy-Focused Firo Cryptocurrency Suffers 51% Attack*, COINTELEGRAPH, Jan. 20, 2021; Carter & Jeng, *supra* note [168], 16.

[174] Carter & Jeng, *supra* note [168].

[175] *See e.g.*, Attah, *supra* note [169].

[176] Avan-Nomayo, *supra* note [173].

[177] Jamie Redman, *Privacy Coin Verge Suffers Third 51% Attack, Analysis Shows 200 Days of XVG Transactions Erased*, BITCOIN.COM, Jul. 2, 2022, https://news.bitcoin.com/privacy-coin-verge-third-51-attack-200-days-xvg-transactions-erased/.

[178] Carter & Jeng, *supra* note [168], 14; David Siegel, *Understanding the DAO Hack*, COINDESK, Mar. 9, 2022.

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Wormhole).[179] Ronin supported the blockchain-based operations of Axie Infinity, a play-to-earn gaming platform with around two million users, and game-based transfers of around $3.6 billion in crypto.[180] Wormhole, in turn, sought to bridge the Ethereum blockchain with that of Solana, popular for hosting gaming apps.[181] As vulnerable add-ons to blockchains, bridges have offered a backdoor into the value held on popular chains, and presented a source of further structural risk attaching to large-scale crypto-plumbing.[182]

2. Trading Costs on Crypto Blockchains

These examples highlight some of the frictions facing users of cryptocurrencies when trading their coins. On one hand, they reflect standard concerns: (i) search costs and limited/unpredictable liquidity; (ii) delays and uncertainties about execution and settlement; (iii) the prospect of fraud, manipulation, and theft of assets. But blockchain-based dynamics also raise novel considerations. For one, traders must gain technological understanding of crypto-infrastructure alongside that of the claim being bought and sold, increasing entry costs.[183] Additionally, the permutations of crypto-specific risks can be unpredictable and lack a detailed market history from which to extrapolate their possible impact. A 51% attack, for example, wiping out a record of seemingly immutable transactions, upends whatever expectation of finality a trader might have had when entering the blockchain. How likely such an attack might be, what it might cost a user, and potential mitigation are questions that cannot be easily answered with still-emerging technology, sporadic disclosure and a limited operational history in the marketplace.

Crucially, traders lack the usual recourse for harms suffered. A standard bilateral trade in securities markets generally comes with contractual mechanisms that can help make parties whole if one of them is in breach. If one side does not perform, the other can sue.[184] Or they might negotiate a solution

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[179] Prashanth Jha, *The Aftermath of Axie Infinity's $650M Ronin Bridge Hack*, COINTELEGRAPH, Apr. 12, 2022; Michael P. Regan & Vildana Hajric, *Inside Jump Trading's Response to a $325 Million Wormhole Heist*, BLOOMBERG, Mar. 22, 2022.

[180] Bianca Britton & Jason Abbruzzese, *'Is My Money Gone?' A Thriving Crypto Community Deals with a $600 Million Theft*, NBC NEWS, Apr. 10, 2022.

[181] Solana is also popular for facilitating trades in non-fungible tokens (NFTs). For analysis of the hack, Jordan Finneseth, *Traders Debate Whether Solana (SOL) is a Buy Now That It's Down 87% From Its All-time High*, COINTELEGRAPH, Jun. 30, 2022; Chinanalysis, *Lessons from the Wormhole Exploit: Smart Contract Vulnerabilities Introduce Risk; Blockchains' Transparency Makes It Hard for Bad Actors to Cash Out*, Feb. 3, 2022, https://blog.chainalysis.com/reports/wormhole-hack-february-2022/.

[182] *See e.g.,* Lily Hay Newman, *Blockchains Have a 'Bridge' Problem, and Hackers Know It*, WIRED, Apr. 3, 2022.

[183] On the significance of understanding technological details see generally, Brummer, *Disclosure, Dapps and Defi*, *supra* note [20] (proposing more accessible disclosure norms).

[184] *See e.g.,* International Swaps & Derivatives Organization, *Legal Guidelines for Smart Derivatives Contracts: the ISDA Master Agreement* (Feb. 2019), 7-12; 32-34.

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based on soft pressure and a repeat relationship. Compensation for theft or fraud might also arise through regulatory intervention or litigation.[185]

On a public blockchain, however, the picture is much less clear-cut. Risk lies with the trader. Failures cannot be remedied, and damages recouped, except under exceptional and *ad hoc* circumstances. Because transactions are supposed to be unchangeable once imprinted on the blockchain, reversal is generally not permitted. Rather, parties must undertake another, separate transaction in order to economically reverse the terms of the deal.[186] Importantly, it is not practicable for a participant to sue another within a pseudonymous system with global reach. Shaming, peer pressure and public reproach that might work within a traditional marketplace are defanged in the context of a pseudonymous blockchain.[187] Distributed public blockchains also raise thorny issues about ascribing responsibility, and whether any actors exist (e.g., developers) that might provide a centralizing locus of responsibility within an otherwise diffuse system.[188] This adds to the importance of self-reliance and means that some may decide that the risk is too much for them.

To be sure, users have, on occasion, received redress following an attack on the system. But the record is *ad hoc*, unsystematic, and exceptional. The DAO hack on the Ethereum blockchain is case-in-point. Owing to its scale, the attack prompted Ethereum's governance to carry out a deep reset in order to negate the effects of the breach. In the Wormhole hack, a prominent backer of the project – Jump Crypto – stepped in to backstop the losses.[189] Axie Infinity's parent company, Sky Mavis, raised $130 million to mitigate the damage from the hack on Ronin.[190] In other cases, the Justice Department has unraveled hacker identities from data on the blockchain and seized stolen crypto-assets (by acquiring private keys to the accounts).[191] Still, these measures are unusual. Moreover, they rely on known actors from outside the blockchain – a fact far removed from Nakamoto's vision.[192]

B. The Essential Role of Centralized Crypto-Exchanges

[185] See e.g., Urška Velikonja, *Public Compensation for Private Harm: Evidence from the SEC's Fair Fund Distributions*, 67 STAN. L. REV. 331 (2015) (detailing funds allocated by the SEC for victims of securities fraud).

[186] Nakamoto, *supra* note [141],

[187] Skeel, *supra* note [106]; Kevin Werbach, *Trust, But Verify: Why the Blockchain Needs the Law*, 33 BERKELEY TECH L. J. 489 (2018) (noting the need for governance for public blockchains).

[188] For discussion on the slippery concept of decentralization see e.g., Angela Walch, *Deconstructing 'Decentralization': Exploring the Core Claim of Crypto Systems*, in Chris Brummer (ed.), Crypto Assets: Legal and Monetary Perspectives (2019).

[189] Regan & Hajric, *supra* note [179].

[190] Jha, *supra* note [179].

[191] *See e.g.*, Katie Benner, *Justice Dept. Seizes $3.6 Billion in Bitcoin and Arrests Married Couple*, N.Y. TIMES, Feb. 8, 2022; Dan Mangan, *World's Biggest Darknet Marketplace, Russia-Linked Hydra Market, Seized and Shut Down, DOJ Says*, CNBC, Apr. 5, 2022.

[192] Carter & Jeng, *supra* note [168], 21-22.

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Exchanges have transformed the architecture and experience of cryptocurrencies. Designed to list digital assets on their platforms, as well as to match buyers and sellers looking to trade them, these firms have captured trading volumes as well as mainstream attention from everyday consumers and Wall Street financial institutions. At the market's heights in October 2021, 419 cryptocurrency exchanges were intermediating $112 billion in trades daily.[193] Many became household names, even before FTX's spectacular collapse made headline news. Taking primetime slots during Superbowl 2022, endorsed by celebrities like Matt Damon and Tom Brady, and lending their brand to sports arenas in Los Angeles and Miami, their rapid ascent into popular consciousness has been swift and visible.[194] Moreover, they have become institutionalized. Coinbase, the third most active platform by volume, is a publicly traded U.S. company.[195] Exchanges like Coinbase and FTX (prior to its collapse) partnered with traditional financial firms like BlackRock, Visa and PNC Bank to offer crypto-linked products.[196] In May 2022, FTX announced that it was looking to expand into U.S. stock markets by offering U.S. customers stock-trading facilities on its app.[197] It also took an equity stake in a regulated national securities exchange, the Investors Exchange (IEX).[198]

The appeal of exchanges makes particular sense in a context otherwise requiring users to interact on blockchains. Like traditional securities exchanges, cryptocurrency platforms can mitigate some (but not all) of the friction endemic to blockchain-based trading. The functionalities offered by trading platforms carry particular value in crypto, facilitating broad uptake of cryptocurrencies across crypto/traditional markets.

1. Search Costs, Trading and Efficiency

Cryptocurrency exchanges alleviate the search costs and limited liquidity experienced on blockchains. By convening buyers and sellers – and matching them using the exchange's intermediation – those seeking a particular trade for their cryptocurrencies can find it more easily. This model stands in

[193] Grandview Research, *Cryptocurrency Exchange Platform Market Size, Share & Trends Analysis Report By Cryptocurrency Type, By End Use, By Region, And Segment Forecasts, 2021 To 2028*, https://www.grandviewresearch.com/industry-analysis/cryptocurrency-exchange-platform-market-report#:~:text=The%20market%20growth%20is%20highly,106%20million%20in%20January%202021..

[194] *See sources and discussion supra* notes [1]- [4].

[195] COINMARKETCAP, TOP CRYPTOCURRENCY SPOT EXCHANGES, https://coinmarketcap.com/rankings/exchanges/; Coinbase Global, Form S-1 (Feb. 25, 2021), https://d18rn0p25nwr6d.cloudfront.net/CIK-0001679788/699359de-d974-4ad9-b7f6-5031f2f432d3.pdf.

[196] Grandview Research, *supra* note [193]; Brush, *supra* note [140].

[197] Alexander Osipovich, *Crypto-exchange FTX Expands into Stock Trading*, WALL ST. J., May 19, 2022.

[198] Michael Bellusci, *The Size of FTX's Investment Was Not Revealed, but an FTX Spokesperson Said It Will Become a "Significant" Shareholder in IEX*, COINDESK, Apr 22, 2022.

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contrast to that of the public blockchain – where the idea of intermediation by a single entity – is conceptually anathema.[199]

But leading exchanges have shown themselves adept at attracting users and building volume.[200] Exchanges provide a way to transact in cryptocurrencies using familiar and accessible ways that feel far removed from the technically daunting environs of blockchains. Sophisticated and user-friendly interfaces (including for smartphones) allow users a quick on-ramp into the exchange as well as the ability to use common tools to pay and engage (e.g., credit/debit cards). In many cases, these interfaces are designed for a curious newcomer, while frequently also including tiers for those that may be more experienced as well as institutional entrants.[201] The dollar amounts needed to sign-up are generally small, promising fractions of crypto for low-dollar value.[202] Exchanges routinely also offer lucrative incentives to those joining, often in the form of money rewards.[203]

Importantly, exchanges must also maintain this volume. Where liquidity is sticky – that is, users become repeat players – exchanges can flaunt a deeper base of liquidity that is continually capable of attracting new traders and lowering transaction costs.[204] However, crypto-exchanges confront challenges in this regard. Users can easily migrate to other venues. Or they may drop out of the crypto market altogether. The aftermath of the FTX collapse – has understandably led to a drop in retail customer participation. According to one industry study, retail trading flows fell from 45% in 2022 to 34% of all trading flows at the start of 2023.[205]That exchanges seek to appeal to experimental newcomers with small amounts of money makes them vulnerable to these entrants leaving quickly. Beginners may lose confidence. They may also put in low amounts that reflect limited loyalty to a particular venue.

To encourage more resilient liquidity and a repeat customer base, however, exchanges deploy sophisticated techniques to incentivize users back to the venue. "Gamification" is one example.[206] For example, exchanges may host competitions for successful traders, set-up league tables, and reward those

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[199] Nakamoto, *supra* note [141], 1-2. Chris Giancarlo, *Let the Market Determine Bitcoin's Value*, FIN. TIMES, FEB. 25, 2021(noting Bitcoin as a social construct, not a government one).

[200] But see, Lin William Cong, Xi Li, Ke Tang & Yang Yang, *Crypto Wash Trading*, Working Paper (Apr. 2021) (showing that wash trading can contribute to reported volume).

[201] See e.g., Jessica Elliott, *Binance vs. Coinbase*, INVESTOPEDIA, https://www.investopedia.com/binance-vs-coinbase-5120852.

[202] See e.g., KRAKEN, BUY BITCOIN & CRYPTO, https:// https://www.kraken.com/en-us/ ("Get started in minutes with as little as $10. Buy Crypto with $10.").

[203] See e.g., COINBASE, SIGN UP, https://www.coinbase.com/signup (offering up to $200 worth of Bitcoin after joining and spinning "the wheel of rewards"); GEMINI, https://www.gemini.com/ ("New US customers who sign up to Gemini will get $7 in ETH after they onboard.").

[204] On the benefits of network externalities, see generally, Madhavan, *supra* note [68].

[205] Patricia Kowsmann, *Bitcoin Prices Rally as Larger Traders Take Charge*, WALL ST. J., FEB. 3, 2023.

[206] To be sure, gamification has been scrutinized as a troubling business practice in the stock market. See for example, Annie Massa & Tracy Alloway, *Robinhood's Role in the 'Gamification' of Investing*, WASH. POST, JUL. 19, 2021.

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that win, including publicity on social media.[207] Exchanges might recognize those that bring business with referrals fees.[208] Social networks are harnessed to encourage memberships, possibly creating greater incentive on the part of new users to stay loyal to one exchange.

Cryptocurrency exchanges also offer a suite of commercial products to their users in addition to core trading services. Like traditional exchanges, crypto-venues have developed a cache of products to strengthen ties to regular users and reinforce their loyalty to the platform. This is exemplified in the practice of exchanges lending money to users to fund trading positions. "Margin trading" is a long-established – and highly regulated – activity in stock trading.[209] Generally borrowing money from their brokers (not an exchange), stock traders can amplify their returns (and magnify losses).[210] Owing to its riskiness, margin lending in stock markets has tight controls on how much can be borrowed and the collateral that must be provided.

In cryptocurrency markets, margin lending similarly provides users with a boost to their bet – and offers stunning returns should things turn out well. At the same time, it also risks rapid collapse and steep margin calls if a trade turns against a borrower.[211] In the absence of regulation, crypto-exchanges hew to a range of practices in determining how much to lend users, and on what terms. Some exchanges offer as much as 100x in leverage, while others limit leverage to lower amounts.[212] Importantly, whereas brokers, not exchanges, supply margin loans in stock markets, exchanges often fund the practice in crypto.[213] Exchanges may also offer credit cards (with rewards in crypto) as well as general loans to customers, promising favorable interest rates and collateralization using crypto at the venue.[214]

In addition to margin lending, loans, and credit cards, exchanges also offer innovative mechanisms to create financial linkages between themselves and the user. Importantly, they routinely offer products that include depositing user funds in outwardly lucrative crypto-ventures. A popular way for exchange

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[207] *See e.g.*, Lujan Odera, *Crypto-exchanges "Gamification" Promoting User Adoption As Bitcoin Halving Nears*, COINGAPE, MAY 11, 2020, https://coingape.com/crypto-exchanges-gamification/.

[208] *See e.g.*, BINANCE, JOIN THE BINANCE AFFILIATES PROGRAM, OCT. 4, 2020, https://www.binance.com/en/blog/community/join-the-binance-affiliate-program-376015737666441216.

[209] Bd. of Governors of the Fed. Rsrv. Sys., Regulation T, 2 CFR §220; Sec. & Exch. Comm'n, *Margin: Borrowing Money to Pay for Stocks*, Apr. 17, 2009.

[210] *See e.g.*, Charles Schwab, *Margin Loans at Schwab*, https://www.schwab.com/margin#:~:text=Margin%20lending%20is%20a%20flexible,power%20to%20purchase%20more%20securities..

[211] SOFI, GUIDE TO MARGIN TRADING CRYPTO, https://www.sofi.com/learn/content/margin-trading-crypto/#:~:text=What%20Is%20Margin%20Trading%20Cryptocurrency,they%20have%20to%20work%20with.

[212] *See e.g.,* CRYPTOVANTAGE, TOP CRYPTO MARGIN TRADING EXCHANGES, https://www.cryptovantage.com/best-crypto-exchanges/best-exchanges-for-margin-trading/.

[213] SOFI, *supra* note [211].

[214] *See e.g.,* COINBASE, BORROW CASH USING BITCOIN AS COLLATERAL, https://www.coinbase.com/borrow; *See e.g.,* CRYPTO.COM, CRYPTO.COM VISA CARD, https://crypto.com/us/cards.

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products to generate returns arises out of forms of "yield farming" – where venues take coins held by them on behalf of users and deposit these in defi or other ventures where they accrue interest/rewards.[215] The returns can then be passed on, in part, to the user. For customers that do not wish to scour crypto ventures for lucrative opportunities, such yield farming provides a profitable convenience.[216] At the same time, it works to interlink customers economically with exchanges and fosters reliance between them for the provision of an ever-greater range of financial products and services.

But even with repeat customers, exchanges still need mechanisms to support liquidity and preserve continuity in trading, as a means of reducing search costs and ensuring that parties can trade whenever they wish. Stock markets have long confronted the problem that trading opportunities can dry up.[217] Without a counterpart on the other side, liquidity dries up, prices become volatile, and the utility of the exchange is damaged.[218]

Whereas traditional markets rely on securities firms to be continually available to trade with investors, crypto-exchanges rely on a mix of strategies that can include using their own capital to trade with users.[219] In other words, venues put their own balance sheet on the line to facilitate trading or "make markets" if a counterparty cannot be found. To be sure, several traditional securities traders have ventured onto crypto-exchanges in a bid to expand their footprint.[220] But crypto markets can also include their own expert players that provide liquidity across platforms.[221] But in certain cases, exchanges themselves (or though related entities) might step in and trade directly with a customer to maintain liquidity.[222] Infamously, FTX deployed the services of Alameda Research, a quantitative trading firm co-founded by Sam Bankman-Fried, that FTX relied on to transact directly with customers as a market maker.[223] Binance.US has been tied to two trading firms, Sigma Chain AG and

[215] *See e.g.,* COINBASE, COINBASE MAKES IT EASY TO EARN YIELD WITH DEFI, https://blog.coinbase.com/coinbase-makes-it-easy-to-earn-yield-with-defi-bd38156e2715.

[216] Gemini offered a program, Gemini Earn, that took crypto deposits and offered yields of between 0.45% and 8%, partnering with Genesis Global Capital, a crypto-broker, for intermediation in generating these returns. Genesis took customer funds and invested these across a range of crypto-enterprises, including lending them out to various firms. Jason Nelson, *Gemini, Genesis Reach $100 Million Agreement Over Earn Program*, Decrypt, Feb. 6, 2023; Matt Levine, *SEC Comes for Gemini Too Late*, BLOOMBERG, JAN. 17, 2023.

[217] See e.g., Lawrence R. Glosten, *Insider Trading, Liquidity, and the Role of the Monopolist Specialist*, 62 J. Bus. 211, 211–14 (1989) (describing the need for designated liquidity suppliers).

[218] Vildana Hajric & Katherine Greifeld, *Bitcoin's Wild Weekends Turn Efficient Market Theory Inside Out*, BLOOMBERG, Jan. 9, 2021.

[219] Eva Szalay, *Battle for Dominance Heats Up in Cryptocurrency Trading*, FIN. TIMES, Jan. 5, 2022.

[220] Szalay, *supra* note [219].

[221] Szalay, *supra* note [219].

[222] Eva Szalay, *Crypto-exchanges' Multiple Roles Raise Conflict Worries*, FIN. TIMES, Jan. 5, 2022; Gary Silverman, *SEC Explores Segregating Businesses at Crypto-exchanges*, Apr. 4, 2022.

[223] Matthew Goldstein et al., *How FTX's Sister Firm Brought the Crypto Exchange Down*, N.Y. TIMES, NOV. 18, 2022.

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Merit Peak, that allegedly act as market makers. Both trading firms are reportedly controlled by Binance's CEO, Changpeng Zhao.[224]

There are several explanations for why a crypto-exchange (or closely related entities like Alameda) might want to act as a market maker on the platform. For one, a venue with unreliable liquidity will struggle to capture and retain business. In addition, certain digital assets may not always attract reliable trading volumes, especially if they are newly listed thinly traded tokens. Importantly, crypto-exchanges routinely operate around the clock, seven days a week, even on public holidays.[225] Unlike traditional markets that complete a day's trading by 4pm (Eastern Time) and are closed on weekends, crypto-exchanges follow a schedule that demands continuous availability of liquidity. This poses problems for exchanges. Institutional players may be less likely to work on the weekends or outside normal trading hours. With banks closed, funding for large crypto trades may also be hard to obtain.[226] Perhaps not surprisingly, the price of Bitcoin experiences some of its most volatile trading and price fluctuations on weekends. According to commentators, this can be explained by the relative poorer levels of liquidity on crypto trading platforms at odd hours, especially within a marketplace consisting of a multiplicity of trading venues.[227]

While an exchange trading with a customer might solve immediate concerns about liquidity, the economic dependence fostered by this practice further ties the fortunes of an exchange with that of its customer. Exchanges can come to rely on their customers for gains from market making, causing the venue to become more beholden to this business.[228] Unsurprisingly, the ability of the exchange to trade against its own customers comes with any number of governance risks. The market maker might seek to "front run" or trade ahead of customers – knowing that a customer is buying Bitcoin, a market maker can buy it first and sell it to the customer at a higher price. The market maker could also get preferential treatment on the exchange. For example, Alameda was permitted private exceptions from risk controls deployed by FTX against other customers. Even while suffering losses, Alameda could continue to trade by borrowing to far higher limits than other customers were allowed. This gave Alameda an obvious edge and exposed the exchange and its customers to the risks of Alameda's outsize footprint as liquidity generator.[229]

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[224] Caitlin Ostroff et al., *SEC Probes Trading Affiliates of Crypto Giant Binance's U.S. Arm*, WALL ST. J., FEB. 15, 2022; Matthew Goldstein, Emily Flitter and David Yaffe-Bellany, *S.E.C. Accuses Binance of Mishandling Funds and Lying to Regulators*, N.Y. TIMES, JUN. 5, 2023.

[225] Ekin Genç, *Is There a 'Best' Time to Trade Crypto? Here's What the Data Says*, COINDESK, Apr. 26, 2022.

[226] Genç, *supra* note [225]; Hajric & Greifeld, *supra* note [218].

[227] Hajric & Greifeld, *supra* note [218].

[228] *See sources cited supra* note [222].

[229] See e.g., Joshua Oliver, *Sam Bankman-Fried's Trading Shop Was Given Special Treatment on FTX for Years,* FIN. TIMES, DEC. 3, 2022; U.S. vs. Sam Bankman-Fried 22 Crim. 673 (Dec. 13, 2022).

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2. Standards, Rules, and Norms

Transacting on the blockchain demands that users accept the necessity of self-reliance. If the transaction does not get picked up and processed, for example, there is little the user can do but try again. In the event of an attack, resulting the user losing value, recourse may depend on the scale of the disruption and the generosity of stakeholders.

As with traditional securities and commodities exchanges, crypto-exchanges generally provide some system of rules and standards for their venue.[230] By offering a predictable and standardizing environment for transferring value, exchanges can better attract those that might be skittish about transacting on pseudonymous blockchains. Three areas stand out as particularly effective for establishing such contractual assurance and organization.

First, crypto-exchanges usually establish entry procedures for users and for listing assets. New users must generally disclose information to verify identity and solvency.[231] Venues might also seek to establish the level of a user's expertise.[232] Prominent exchanges highlight this onboarding as important for ensuring compliance with money laundering and terrorist financing regulations.[233] These entry procedures distinguish exchanges from public blockchains. Users should be clearly identified (to the exchange), and their conduct can thus be monitored, scrutinized, and sanctioned.

Secondly, crypto-exchanges also tend to set out detailed procedures to govern trading. These private rules regulate aspects of how a trade must be operationalized from idea to execution. Behaviors like manipulation, spoofing, wash trading and fraud, are contractually prohibited.[234] This direct oversight of the trading process stands in contrast to the distributed verification on public blockchains, allowing exchanges to offer some comforts of an organized and streamlined trading ecosystem.[235]

Relatedly, unlike public blockchains, many exchanges generally provide mechanisms for dealing with breakdowns on the platform. Whereas

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[230] *See e.g.*, COINBASE, COINBASE USER AGREEMENT, https://www.coinbase.com/legal/user_agreement/united_states; CRYPTO.COM, CRYPTO.COM APP T&CS (MAY 2022), https://crypto.com/document/mco_services; GEMINI, USER AGREEMENT, TIMES, https://www.gemini.com/legal/user-agreement#section-using-our-services; KRAKEN, TERMS OF SERVICE, https://www.kraken.com/en-us/legal;

[231] *See e.g.*, GEMINI, USER AGREEMENT, *supra* note [230].

[232] Elliott, *supra* note [201].

[233] On sanctions compliance in the context of the invasion of Ukraine, see e.g., Hannah Lang, *Crypto Industry on Defensive as Ukraine Crisis Spotlights Russia Sanctions Compliance*, REUTERS, MAR. 10, 2022; Ryan Lawler, *Crypto-exchanges Resist Calls to Block Russian Users,* AXIOS, MAR. 1, 2022 (noting that exchanges resisted blocking all Russian addresses on their platforms, but only those on the sanction lists); BLOOMBERG, *Russia's $124 Billion Crypto Market Is Pinched by Ukraine Sanctions*, Apr. 21, 2022 (noting Binance limiting access to large Russain accounts).

[234] *See e.g.*, COINBASE, COINBASE MARKETS TRADING RULES, https://www.coinbase.com/legal/trading_rules.

[235] Public blockchains can adhere to different models of consensus. See, Irresberger et al., *supra* note [166]. They may ultimately be governed by a more central layer. Walch, *supra* note [188].

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decentralized blockchains seek to operate without intermediaries and ask users to bear the risks of failure, crypto-exchanges usually provide some recourse if the venue fails.[236] A number of formal and informal mechanisms have emerged to aid customers seeking redress. Formally, terms of service generally include contractual provisions for resolving disputes between the user and exchange, including resolution through arbitration.[237] Helplines are also available.[238] Users might have rights to specific funds allocated to compensate them for damage suffered on account of hacks and attacks on exchange accounts.[239] More informally, social media offers a convening forum for users to share grievances. Leading exchanges often maintain high visibility on social media, with founders and key officers regularly communicating publicly.[240] This offers users another strategy through which to express preferences informally. Breakdowns in service routinely prompt furor on popular social media websites (e.g., Reddit).[241] To be sure, these formal and informal resolution mechanisms may not always work as advertised. Comments on social media routinely speak to the frustrating experiences of users.[242] Even for those seeking formal recourse, attempts may be stymied by procedural difficulties in launching a lawsuit.[243] Nevertheless, albeit imperfect, such measures offer a clear contrast between the self-reliance expected of users by public blockchains and attempts to create a more formally responsive environment within crypto-exchanges.

Thirdly, like NYSE and Nasdaq are for public companies, crypto-exchanges have emerged as key players for listing new digital assets for trading, acting as gatekeepers to screen out scams and dubious ventures.[244]

This vetting process for new listings gives exchanges a prized perch from which to view the crypto ecosystem. By dint of this function, they assume a critical role in interrogating the technology, governance, and value proposition underlying emerging coins. Beyond listing established assets like Bitcoin or Ether, exchanges have taken on a role that tasks them with investigating coins

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[236] Nick Wells & Eric Chemi, *A Short History of Stock Market Glitches,* CNBC, Aug. 22, 2010.

[237] *See e.g.*, COINBASE, COINBASE USER AGREEMENT, APPENDIX 5: ARBITRATION AGREEMENT, https://www.coinbase.com/legal/user_agreement/united_states#appendix-5-arbitration-agreement.

[238] *See e.g.*, COINBASE, HELP, https://help.coinbase.com/en.

[239] *See e.g.*, Ben Bartenstein, *Binance Builds Up $1 Billion Insurance Fund Amid Crypto Hacks,* BLOOMBERG, Jan. 31, 2022.

[240] *See e.g.*, TWITTER, @SBF, https://twitter.com/SBF_FTX?ref_src=twsrc%5Egoogle%7Ctwcamp%5Eserp%7Ctwgr%5Eauthor (Account of Sam Bankman-Fried, CEO of FTX); TWITTER, @CZ_BINANCE, https://twitter.com/cz_binance?ref_src=twsrc%5Egoogle%7Ctwcamp%5Eserp%7Ctwgr%5Eauthor (Account of Changpeng Zhao, CEO of Binance).

[241] *See e.g.*, REDDIT, R/COINBASE, https://www.reddit.com/r/CoinBase/.

[242] *See e.g.*, REDDIT, R/COINBASE: WHY IS COINBASE DOWN, https://www.reddit.com/r/CoinBase/comments/qh4kez/why_is_coinbase_down/.

[243] *See e.g.*, Russell Brandom, *Binance is Facing an Unconventional New Legal Claim Over May 19th Outage*, THEVERGE, Aug. 19, 2021 (highlighting difficulties in commencing legal process against Binance on account of its lack of headquarters).

[244] *See e.g.*, COINBASE, COINBASE, A GUIDE TO LISTING ASSETS ON COINBASE, https://blog.coinbase.com/a-guide-to-listings-assets-on-coinbase-2556fee4efa6; BINANCE, BINANCE LISTING TIPS FROM CZ, https://www.binance.com/en/blog/from-cz/binance-listing-tips-from-cz-421499824684902218.

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native to emerging blockchains and blockchain applications. Stated differently, they represent a centralizing locus of scrutiny for assets that are created and exist within a more decentralized, diffuse environment where mechanisms to vet, audit and stress-test are typically less systematic (by design). Notably, defi seeks to create applications that aim to operate in an automated way, reliant on code, rather than an intermediary. In a defi lending app, for example, users borrow and lend coins by depositing them in a smart program, whose algorithm automatically works out the rate of interest, collateralization, and the risk thresholds at which the debt should be called (or liquidated, meaning the collateral is seized and sold.) In 2022, around $50 billion was deposited in defi lending protocols, growing from essentially nothing just two years prior.[245] Like Bitcoin, defi applications often reward those that protect networks, allocating coins to those that use the system and have a stake in promoting its safe workings. Compound, a defi lending application, issues governance tokens that allow customers to take decisions about its operations. Though this token is native to defi, minted by an application that claims to lack a central decision-making body, it is an exchange-listed asset that can be widely traded.[246]

Exchanges ought to have incentives to carefully vet new coins prior to listing. As a logistical matter, each new coin's code must be incorporated into an exchange's own internal listing and trading systems, meaning that a venue's customer accounts (or wallets) must be able to recognize and accept as value the input represented by the code of a listed coin. Where integration fails, an exchange can suffer expensive problems in executing transactions against that coin and reflecting outcomes in user accounts.[247] Further, the coin's code may have bugs that expose the exchange's systems to risk.[248] The underlying blockchain for a project may be susceptible to attacks, leaving the exchange's holdings in the coin vulnerable to being wiped out.

Failure to vet new listings also endangers the reputation of an exchange. Collapsing crypto ventures – and problem coins – create sizable risks for a platform. For one, where a listed coin and its blockchain collapse, exchange customers are likely to panic-sell their holdings. High volumes of users flooding into the exchange will put pressure on its available liquidity, triggering a cash crunch for the venue if users wish to sell *en masse* and extract their money. When an exchange cannot meet these demands, it may have to consider restrictions on the ability of users to trade or exit, risking serious reputational damage and subsequent firestorms on social media.

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[245] Sirio Aramonte, Sebastian Doerr, Wenqian Huang & Andreas Schrimpf, *Defi Lending: Intermediation without Information*, BIS Bulletin Number 57, 1-4 (Jun. 14, 2022)

[246] *See e.g.*, COINBASE, COMPOUND, https://www.coinbase.com/price/compound. On compound and its claims to decentralization, GEMINI, WHAT IN COMPOUND IN 5 MINUTES, https://www.gemini.com/cryptopedia/what-is-compound-and-how-does-it-work. On identifiable actors working as part of seemingly decentralized firms, Andrew Thurman, *Compound Founder Says $80M Bug Presents 'Moral Dilemma' for DeFi Users*, COINDESK, OCT. 1, 2021.

[247] *See e.g.*, COINCRUNCH, *Binance to Re-build Doge Wallet After Glitch*, Nov. 24, 2021.

[248] *See sources cited supra* note [244].

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The May 2022 crypto-crisis made this connection clear. The downward spiral of the $58 billion TerraUSD/Luna "algorithmic" stablecoin system extracted a heavy price from leading exchanges.[249] This stablecoin – designed to maintain a 1:1 peg with the US dollar, where one coin would always be worth one dollar – was underpinned by a mathematical program that maintained the peg using automated changes to supply and demand of available coins. Relying on mathematics rather than real-world reserves of assets (like cash) to pay out on customer claims, the Terra/Luna stablecoin ecosystem unraveled rapidly as confidence in its workings plunged.[250] On the day that TerraUSD/Luna's price crashed by 96%, Coinbase reported major outages. Users noted shutdowns in access, difficulties completing transactions and in withdrawing funds.[251] When the defi lending platform Celsius collapsed in mid-June 2022, the price of its native coin (CEL) crashed, losing 70% of its value within an hour. Binance's BNB coin experienced a 9% loss on the news and Coinbase's share price fell the following day from $60 per share on June 10 to $46 on June 13.[252]

These dynamics underscore the broader point that the fortunes of cryptocurrency exchanges are closely tied to the health of the market, giving venues existential skin-in-the-game to vet crypto-projects and their coins.[253] In May 2022, Coinbase reported a quarterly loss of $430 million and two million users, with the firm's share price lower by 85% from an all-time high in November 2021.[254] Leading exchanges like Coinbase, Gemini and Crypto.com instituted lay-offs and cost-cutting.[255] Dollar trading volumes on exchanges fell rapidly from late 2021, falling from a high of around $6.1 trillion in November 2021 to $3.8 trillion at the end of the first quarter in 2022. Over this period, the capitalization of cryptocurrencies lost around two trillion dollars in value.[256]

3. Certainty, Speed, Convenience

[249] On Terra/Luna's history prior to the collapse, Alexander Osipovich, *Cutting-Edge Crypto Coins Tout Stability. Critics Call Them Dangerous*, WALL ST. J., Apr. 18, 2022; Alexander Osipovich & Jiyoung Sohn, Do Kwon's *Crypto Empire Fell in a $40 Billion Crash. He's Got a New Coin for You*, WALL ST. J., Jun. 22, 2022 (detailing the scale of the Terra blockchain and Terraform Labs).

[250] Osipovich & Sohn, *supra* note [249].

[251] Krisztian Sandor & Ekin Genç *The Fall of Terra: A Timeline of the Meteoric Rise and Crash of UST and LUNA*, COINDESK, Jun. 1, 2022; Russell Brandom, *Coinbase Experiencing Major Outage as Crypto Prices Plummet*, THEVERGE, May 12, 2022.

[252] Daniel Roberts, *Celsius Tanks 70% in 1 Hour After Company Pauses Withdrawals to 'Stabilize Liquidity,'* DECRYPT, JUN.12, 2022; On Coinbase's share price for June 13, 2022, YAHOO FINANCE, COINBASE GLOBAL: HISTORICAL DATA (JUNE 2022).

[253] *See e.g.*, COINMARKETCAP, CELSIUS, https://coinmarketcap.com/currencies/celsius/.

[254] David Yaffe-Bellany, Erin Griffith & Ephrat Livni, *Cryptocurrencies Melt Down in a 'Perfect Storm' of Fear and Panic*, N.Y. TIMES. May 12, 2022.

[255] Emily Nicolle & Olga Kharif, *A $2 Trillion Free-Fall Rattles Crypto to the Core*, BLOOMBERG, Jun. 26, 2022.

[256] Nicolle & Kharif, *supra* note [255].

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Exchanges can offer quick, cheap, and reliable execution of trades. By promising near instant execution and settlement of trades, platforms can satisfy users looking for greater certainty and convenience in trading than they might find on public blockchains.

Public blockchains deploy distributed consensus to verify the authenticity of the transactions. This can cost time, especially during periods of high traffic. It can also create uncertainty if miners fail to pick up and process trades. Even if public blockchains rely on immutability, with the ledger constituting a definitive and transparent record, finality still tends to be probabilistic. It depends on whether transactions may be unwound, and blocks rewritten, such that cautious users might wish to wait until their transaction is deeply embedded into the history of the chain before feeling confident that their trade is completely settled.[257]

By contrast, crypto platforms promise reliable, rapid finality and hew to a radically different model for settling trades. Rather than settling on the blockchain, most crypto-exchanges settle their trades on the books of the exchange itself.[258] Because of this, settlement is virtually instantaneous, able to move value between user accounts as soon as transactions become binding. In other words, crypto-exchanges entirely internalize the settlement process. By recording trade outcomes on their own ledger, rather than on the blockchain, exchange settlement can occur "off-chain."[259] This helps to effectively guarantee settlement and create the potential for swift, low-cost, and predictable movement of value between users.

Choosing this mode of settlement has three consequences. First, to move value between users, exchanges generally seek out authority to access these accounts. Whereas blockchains demand that users rely on their own resources to look after private passwords, exchanges commonly request that users give custody of customer assets and access to these funds to the venue. When the user opens their account with the exchange and transfers cash and crypto to exchange wallets, the exchange also requests custody of a customer's wallet keys and control over how this value is stored. As Adam Levitin observes, these custody arrangements often give exchanges enormous leeway, such that some might mix customer money into an omnibus account.[260] Exchanges can offer variation on these arrangements. Some customers may contract with an exchange for a segregated account.[261] Or, an exchange may

[257] Gauba, *supra* note [155].

[258] Johnson, *supra* note [145], 1921-1922 (noting the impediments of "on-chain" settlement).

[259] *See e.g.*, COINBASE, COINBASE MARKETS TRADING RULES, https://www.coinbase.com/legal/trading_rules (noting instantaneous settlement of trades as soon as these are matched on the platform).

[260] Adam Levitin, *Not Your Keys, Not Your Coins: Unpriced Risk in Cryptocurrency*, TEXAS L. REV. 4-5, 14-16 (*forthcoming*) (also noting that legal notions of ownership strain when applied to cryptocurrency custody arrangements).

[261] Levitin, *supra* note [260], 14-15.

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offer customers the option of settling "on-chain," where balances are transferred off the exchange to the customer's account on and using the blockchain.[262]

Secondly, with rapid off-chain settlement and custody of user funds, cryptocurrency exchanges constitute highly consolidated trading institutions, far more so than traditional exchanges. As set out in Part II, securities and commodities exchanges rely on a number of different institutions in the trading process. Once buyers and sellers enter a contract on the exchange, clearinghouses step in to ensure that value moves between the parties. Third party custodians will also be involved to warehouse and protect the assets (e.g., commodities). Brokers might provide financing to customers. By contrast, cryptocurrency exchanges internalize all of these functions within a single firm.[263] Rather than sharing costs with others, crypto-exchanges tend to assume the risk of completing each of these complex tasks themselves, meaning that exchanges can routinely offer customers trading, clearing, settlement, custody, market-making, financing within the course of a single trade. In addition, as noted earlier, exchanges also engage traders with various commercial product offerings, like loans or opportunities for yield farming.

C. Why Exchange Self-Regulation Has Failed (So Far)

Despite their significance, cryptocurrency exchanges have grown up without a system of formal federal public oversight or private industry self-regulation. Lacking mandatory guardrails, exchanges have shown themselves vulnerable to risky experimentation in institutional set-up as well reckless and disruptive behavior (e.g., FTX). Importantly, without a mandate to oversee their venue in accordance with agreed-upon standards, exchanges have, in practical terms, exhibited extremely patchy commitment to safety and norms.[264] To be sure, traditional exchanges, too, have suffered problems. But as discussed below, unique features of the cryptocurrency market – their global and mobile nature, for example – have weakened incentives for venues to voluntarily take on the costs of private industry self-regulation.

First, venues have faced little top-down pressure to develop mechanisms for private oversight.[265] Crypto-exchanges are subject to discrete aspects of federal regulation, notably in relation to money transmission, anti-

[262] *See e.g.*, COINBASE, ON-CHAIN VS. OFF-CHAIN TRANSACTIONS ON COINBASE PRIME, https://help.coinbase.com/en/prime/trading-and-funding/on-and-off-chain-transactions-on-coinbase-prime.

[263] Levitin, *supra* note [260], 14-15.

[264] Johnson, *supra* note [145], 1915-1916.

[265] The SEC has proposed a new framework for regulating alternative trading platforms. Crypto-exchange providers have suggested that it may encompass their platforms; Kia Kokalitcheva, *Crypto Industry Takes on SEC's Proposed New "Exchange" Definition*, AXIOS, Apr. 21, 2022; Johnson, *supra* note [145], 1915-1916.

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money laundering, and terrorist financing.[266] They may also opt-into certain state-based regimes, with New York and Wyoming offering a licensing framework for virtual currency businesses.[267] However, even within specialized state regulatory regimes, there is little priority on tasking exchanges with mandatory self-regulation of their markets.[268]

As the origin story for traditional exchanges shows, however, this absence of federal regulation does not have to doom industry self-regulation. Indeed, it can catalyze it.[269] But the structure of the cryptocurrency industry raises costly hurdles that impede the creation of common standards of conduct. Unlike early attempts within securities markets to develop self-governance, crypto-exchanges have failed at this task so far.

As a starting point, the absence of systematic public regulation has spawned varied experimentation in venue design by dint of relatively low barriers to entry and limited constraints on institutional set-up.[270] Cryptocurrency exchanges number in the hundreds globally, with some being little more than websites while others showcase sophisticated, large-scale operations.[271] Logistically, this has increased the challenges entailed in exchanges sharing information with a view to understanding common risks and developing responses. Rather than a small number of similar actors engaged in a repeat game, a multiplicity of venues and their varying models raise the difficulty involved in coordination, information pooling, analysis, and consensus building. Even among major U.S. crypto-exchanges, venues differ in how they present their business, company ethos, customer base, distinguishing themselves through such features as fees, products, and security. Trading volumes vary widely – as does regulatory context.[272] Some venues opt into a state-based regulatory regime, such as that of New York state. Others remain outside of it.

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[266] Levitin, *supra* note [260], 8-9, 62-64.

[267] Levitin, *supra* note [260], 8-9, 62-66; In New York, a Bitlicense may be obtained for companies that engage in trading functions (e.g., custody) of virtual currencies. 23 N.Y.C.R.R. §§ 200.2(q), 200.3. In Wyoming, Special Purpose Depository Institutions offer a framework for custody of virtual assets; Wyoming Div. of Banking, Special Purpose Depository Institutions, at https://wyomingbankingdivision.wyo.gov/banks-and-trust-companies/special-purposedepository-institutions. Levitin also identifies potential consumer protection rules that could be applicable. For discussion, Levitin, *supra* note [260], 65-67.

[268] For discussion, Levitin, *supra* note [260], 62-64.

[269] Georgy Egorov & Bard Harstad, *Private Politics & Public Regulation*, 84 REV. ECON. STUD. 1652, 1652–57 (2017) (developing a model to show how firms may act to prevent a regulator from acting).

[270] Johnson, *supra* note [145], 1915-1916.

[271] Johnson, *supra* note [145], 1915-1916 (noting failures and fraud across a largely unregulated space); COINMARKETCAP, *supra* note [1], https://coinmarketcap.com/rankings/exchanges/ (visited Jul. 7, 2022); Various websites discuss how to set up a crypto-exchange and note that steps need not be arduous, QUORA, *How Do You Set Up Your Own Cryptocurrency Exchange*, https://www.quora.com/How-do-you-set-up-your-own-cryptocurrency-exchange; Yaroslav Titenok, *How to Build a Cryptocurrency Exchange like Binance,* https://sloboda-studio.com/blog/building-a-cryptocurrency-exchange-like-binance/ (May 23, 2022).

[272] COINMARKETCAP, *supra* note [1]; Jacob Wade, *Kraken vs. Coinbase*, THEBALANCE, https://www.thebalance.com/kraken-vs-coinbase-5119424.

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With this volume and variety, information sharing and coordination between venues is far from straight-forward. A small exchange making a bid for market share might be wary of taking on the costs of large-scale data collection, analysis, and sharing.[273] Crucially, it may not gain much from its efforts. Information gleaned from larger players may not be relevant for the new business. It may also be difficult to process operationally.[274] Data sharing could further result in subjectively unpleasant consequences, like new risk controls, that a small firm is poorly equipped to bear. Even large venues may not want to share their data. Those with vast operations will house rich troves of information. They may not feel like they need data from smaller platforms, viewed as lacking relevance for bigger venues that have long passed the teething stage in their growth cycle.

Such problems are not new.[275] But they are exacerbated within the cryptocurrency industry owing to fierce competition where venues can emerge quickly to jostle for market share. Sharing information can offer gains where it is undertaken strategically, for example, with a view to receiving information from other firms in return.[276] However, data pooling can also be costly. Importantly, it can divulge information about unsavory tactics that firms may be using to juice their business and elbow ahead. With limited regulation, behaviors that may be punished in traditional markets can escape sanction in crypto. Information on aspects like aggressive customer marketing, risky product offerings (e.g., trading loans), fake trading volumes, or tolerance for criminality could result in reputational loss and diminished trust in the market.

For example, crypto-exchanges have shown a tendency to report inflated trading volumes. In one study, scholars highlight divergence in this respect between more and lesser-regulated exchanges. Unregulated venues in their sample inflated trading volumes by an average of around 70%. Platforms reported transaction figures that included trades where customers were trading with themselves – so-called wash trading – that conveyed a fictitious gauge of the venue's popularity. The study observed that these puffed-up disclosures resulted in the platform gaining in rankings and prominence. Misreporting, they show, is more prevalent among newer exchanges with a smaller base of users.[277] From this standpoint, information gathering, and coordination is unhelpful when the data being collected is unreliable and venues have little incentive to correct for misinformation if the competitive gains of opacity are higher.

Again, these features of cryptocurrency markets might not be especially novel: financial markets routinely feature competitive players that still manage

[273] *See e.g.*, Priscilla Koepke, *Cybersecurity Information Sharing Incentives and Barriers*, MIT Working Paper CISL# 2017-13, 3-5 (Jun. 2017).

[274] *See e.g.*, Priscilla Koepke, *supra* note [273], 3-5).

[275] Koepke, *supra* note [273], 3-5.

[276] *See e.g.*, Esther Gal-Or & Anindya Ghose, *The Economic Incentives for Sharing Security Information*, 16 INFORMATION SYSTEMS RESEARCH 186 (2005) (identifying the benefits of sharing information in markets where product substitutability is higher).

[277] Cong et al., *supra* note [200].

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to deploy self-governance (e.g., broker-dealers). But crypto-exchanges suffer from further structural deficits that limit their ability to coordinate. Low barriers to entry and few hard constraints on conduct reduce the chances of building trust within the group. Exchanges – even leading ones – are relatively new. Binance – the largest global venue by volume, for example – was established in 2017.[278] This young, moving cast of players –globally dispersed, subject to varying operating styles and degrees of regulation – impede the creation of stable, reliable bench of firms that can grow trust and social capital through repeated interactions and mutual risk-and-benefit sharing.[279] Exchanges come and go. Even prominent ones, like FTX, can essentially disappear overnight. To take another example, Quadriga – once, Canada's largest crypto trading venue – cratered after its founder died in mysterious circumstances in sole possession of the private keys to customer wallets holding around $125 million in assets.[280] This fragmentation means that alliances that might build trust, cohesion and mutual reliance cannot be easily formed, if they can be formed at all.

These difficulties create fatal setbacks to the coordination game for setting up viable self-regulation where venues cooperatively pursue a collective benefit, forgoing opportunistic, self-interested risk-taking.[281] Not being similarly situated, the benefits to each venue will be uncertain and varied. A smaller platform that can win market share by being aggressive (e.g., marketing fake trading volumes) will face a different trade-off than a successful venue with a loyal customer base. Already benefitting from positive network effects, the established venue may draw more business by showcasing compliance. It can reinforce trust with existing customers as well as attract new business by signaling commitment to high standards. By contrast, a newer platform will have to deploy strategies that can peel away customers from existing venues as well as attract new ones. This pressure can encourage a newer player to defect more in order to differentiate itself, especially if there are few public rules to prevent risky behavior, and where peer discipline is unlikely to impose sanction.

Conversely, larger exchanges that have grown commercial muscle by deploying problematic practices may be wary of changing course. Where greater transparency might reveal insights into risky behavior (e.g., FTX relying on Alameda for market-making absent fair risk controls), commitment to collective discipline may be lacking, or attaching to fairly selective and

[278] Patricia Kowsmann & Caitlin Ostroff, *$76 Billion a Day: How Binance Became the World's Biggest Crypto-exchange*, WALL ST. J., NOV. 11, 2021.

[279] *See e.g.*, Julian M.Müller, Johannes Veile & Kai-Ingo Voight, *Prerequisites and Incentives for Digital Information Sharing in Industry 4.0 – An International Comparison across Data Types,* 148 COMPUTER & INDUSTRIAL ENGINEERING (2020).

[280] Doug Alexander, *Quadriga Downfall Stemmed from Founder's Fraud, Regulators Find*, BLOOMBERG, JUN. 11, 2020.

[281] Bryan Skyrms, *The Stag Hunt*, Presidential Address to the Pacific Division of the American Philosophical Association (Mar. 2001) (detailing the stag-hunt or coordination game, whereby hunters have two options (i) to collaborate and hunt a stag, garnering a bigger collective reward; or (ii) to defect from the collective goal and pursue lesser, but individually rewarding hares. The game has two equilibria – the first, where hunters capture the stag; and the second, where they all chase hares.

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superficial aspects of industry governance to which the larger venue may be able to adhere. Without buy-in from a major player, other top venues may have little incentive to join a system of peer-supervision and collective constraint.

As detailed above, parties also lack information sharing mechanisms and incentives that might help build communication about shared goals. Deficits in information sharing can put paid to efforts even among similarly situated firms that might want to coordinate and self-regulate.[282] As a starting point, venues will give up information when they believe that it will yield a greater reward. This may mean that they volunteer only such data as is cheap to collect, provides basic insights, or is done only in such areas as there is a considered need for real co-operation (e.g., cybersecurity). Rather than being a global endeavor designed to fully map out risks that venues might create for one another and the market, information transfer references specific objectives or minimally intensive efforts. Importantly, even if larger venues do collaborate to share information and facilitate cooperation, they may still choose to act in ways that retain for each the option to take large risks. In any event, if newer/smaller/riskier venues are not on board, with only a handful of venues agreeing to cooperate, the market is still constantly exposed to the risks created by non-collaborative players. This is likely to sharpen the instinct of more collaborative players to maintain for themselves options to take risks, seek private gains and maintain pole position.

It follows that there has been little movement to develop self-regulatory tools within cryptocurrency markets. Indeed, even prior to FTX, exchanges have routinely been implicated in institutionalizing risky governance norms as well as been at the center of frauds, hacks, and thefts.

As detailed earlier, limited consensus on cooperation and few regulatory fetters on firm design mean that crypto-exchanges have pursued risky choices in how they structure their operations. For example, leading exchange groups also comprise venture arms as part of their organization. They use firm capital to fund start-ups and new ventures. While venture funding can result in spectacular returns, it can also create situations where exchanges suffer heavy losses when the investment fails. These activities have come to the fore during the 2022 cryptocurrency crisis. Binance, for example, lost over one billion dollars from its investment in the TerraUSD/Luna stablecoin.[283] As the crypto-market stumbled in May 2022, FTX and its venture arm pursued a series of deals to buy and lend to businesses on the brink of collapse. Committing around one billion dollars, FTX capitalized a diverse range of struggling firms

[282] *See e.g.*, Joseph Farrell, *Cheap Talk, Coordination, and Entry*, 18 RAND J. ECON. 34, 39-40 (detailing that information, even if fully shared, can ever overcome the competition and conflict inhering within a coordination game). On the role of the law in mediating coordination and conflict, see generally, DOUGLAS BAIRD, ROBERT GERTNER & RANDALL PICKER, GAME THEORY & THE LAW (1998).

[283] Stacy Elliott, *Binance CEO Says Exchange Never Sold Its 15.4M LUNA- Now Worth Just $2,461*, DECRYPT, MAY. 16, 2022.

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in a bid to take discounted but risky positions.[284] As FTX itself unraveled a few months later, these bets were revealed as having likely involved use of customer money – and made during a time when FTX itself was suffering heavy losses from the Terra/Luna collapse.[285] Traditional exchanges, too, offer lucrative services and products. But cryptocurrency exchanges have failed to create consensus around what kinds of safeguards might protect their firm from the risks of mixing business lines and risky entrepreneurship, while also ensuring transparency such that external scrutiny and peer discipline may be exercised. Financial exchanges are subject to a host of mandatory protections when things go wrong (e.g., thick capital cushions). Crypto-exchanges, however, have incorporated riskiness into their firm business models, without clarity about how to mitigate the dangers involved. A lack of standardized governance in cryptocurrency markets, then, mean that venues do not face a form of peer sanction for falling short. Venues might pay for harms occurring on their venue. But they are not penalized for violating industry safeguards.

In summary, cryptocurrency exchanges have been essential to the creation of a vibrant mass-market for digital assets. By offering a fix to the frictions of blockchain-based trading, exchanges provide users with efficiencies common to traditional marketplaces. Structurally, they are well-placed to offer industry self-regulation as SROs for crypto-markets. They convene high volumes of users. They also connect into the defi ecosystem by listing coins and gaining exposure to underlying blockchains. From the industry standpoint, they ought to have strong incentives to step up as private regulators. That said, by themselves, exchanges have failed to privately mobilize in offering industry oversight. High competition and a changing cast of venues limit cooperation and information sharing. Conditions for a repeat game to build trust and common social capital has failed to materialize. Without a mandate, the cost-benefit trade-off has not pushed for the development of industry self-regulation despite the gains it can bring for the marketplace and exchanges themselves.

IV. TOWARD CRYPTO MARKET OVERSIGHT

This Part outlines a proposal to introduce a mandate for exchange self-regulation in cryptocurrency markets.[286] A mandate brings exchanges into the regulatory perimeter. As with traditional exchanges like NYSE or Nasdaq, it would require crypto-exchanges to write rules, monitor, and discipline. This move brings crypto-exchanges into line with a long-established model of

[284] Notably, FTX sought to extend liquidity to Voyager Digital and BlockFi, both of which were in a distressed state in summer 2022 and were supposedly rescued by FTX. Hannah Miller & Olga Kharif, *Sam Bankman-Fried Turns $2 Trillion Crypto Rout into Buying Opportunity*, BLOOMBERG, JUL. 20, 2022.

[285] *See e.g.*, Andrew Chow, *Where Did FTX's Missing $8 Billion Go? Crypto Investigators Offer New Clues*, TIME, DEC. 21, 2022; Matt Levine, *FTX's BlockFi Rescue Didn't Work*, BLOOMBERG, NOV. 28, 2022.

[286] On self-regulation for financial markets see generally, Omarova, supra note [22].

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oversight in financial markets. Importantly, by formally coopting exchanges into the role of monitors, cryptocurrency exchanges have to first show that they are safe, well-governed, and capable of exercising supervision and discipline.[287] The fact of requiring designation assures that exchanges face public regulation that can help put exchanges on a safer footing. Exchange incentives to defect from governance norms become less compelling. Reputational expectations are heightened. Public scrutiny risks damages where private oversight fall short.

Crucially, this proposal is not a substitute for public regulation of crypto-markets. It marks a first, urgent step to bring oversight to a market that has shown itself to be vulnerable to outsize risk-taking, opacity, a lack of care for customer interests, and systemic failures, where the collapse of a large firm undermines others. Private self-regulation is also not a panacea, with exchanges still liable to suffer deficits in information, conflicts of interest, and temptations to defect for large gains. However, relying on this tried-and-tested mechanism of industry self-regulation offers a foundation for building a higher quality cryptocurrency market, supporting public regulators, while also forcing industry to take on costs for its own protection.

A. A Public Mandate for Private Self-Regulation

A starting point for creating a layer oversight of cryptocurrency markets lies in designating crypto-exchanges as self-regulatory organizations (SROs).

1. The Public Mandate

As noted in Part III, crypto-exchanges have failed to develop a framework for self-regulation by themselves. In trading, order and integrity can promote liquidity. But venues also gain by behaving in privately profitable but publicly detrimental ways, for example, by touting fake trading volumes or maintaining opacity.

The introduction of a mandate changes the equation. By creating a penalty for failing to perform sound oversight, the gains from private risk-taking have to be higher to be justifiable. Private self-interest and the public good are more *in sync* because it is beneficial for an exchange to engage in rule-making and surveillance. Exchanges also enjoy tangible gains from SRO status. As detailed in Part II, traditional exchanges benefit from legal insulation against suits, prestige, and formal dialogue with regulators. The benefits of bad

[287] On the contentious and often unproductive relationship between the digital asset industry and regulators see generally, Yuliya Guseva, *When the Means Undermine the End: The Leviathan of Securities Law and Enforcement in Digital-Asset Markets*, Working Paper (Sept. 2021).

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behavior, then – reporting fake volumes, conflicts of interest, shunning collaboration with peers, pushing exploitative marketing to customers and looking to front-run them in trading – have to be worth more than the gains available from engaging in oversight.

2. Key Features of the Mandate

This design envisions an approach similar to that taken for more traditional equity markets, with venues wishing to provide major intermediation for digital assets to register and commit to engaging as an SRO. In equities, platforms are divided broadly into two tiers: (i) those that wish to be fully designated a national exchanges; and (ii) smaller venues, transacting in lower volumes that can qualify as alternative trading systems or ATS.[288] Full exchanges represent the major pillars of the equity markets, capable of participating in the national market and competitively capturing orders from any of the participating exchanges so long as they are able to offer the best price.[289] ATS generally offer more bespoke, off-exchange trading environments (e.g., for large block trades), subject to far fewer obligations, but also restricted in their ability to grow.[290] National exchanges are charged with proactively providing rulemaking, oversight and discipline across the marketplace as holders of SRO status. By contrast, ATS do not come under such expansive obligations. But they do still come under some basic requirements – e.g., to ensure fair access to their venue if they happen to trade larger volumes, disclose aspects of their operations and functions to their users and the public, and to maintain the integrity of confidential information.[291] In addition to regularly reporting to the SEC, ATS also supply disclosure to private industry SROs.[292]

A similar categorization would benefit the market for digital asset intermediation. Larger venues seeking out maximum presence and opportunity would apply to be registered as an SRO. This would require them to come under an obligation to proactively look after the market, monitor users, and engage in rule-writing and discipline. Smaller venues, or perhaps less centralized ones, might apply for a lighter designation, an ATS for digital assets. This would reduce the full oversight burden on the alternative venue, but still subject the

[288] Regulation ATS—Alternative Trading Systems, 17 C.F.R. § 242.300-304.

[289] *See* Regulation National Market System Rule 611, Order Protection Rule, 17 CFR 242.611 (2005).

[290] Sec & Exch. Comm'n, *Responses to Frequently Asked Questions Concerning Rule 301(b)(5) under Regulation ATS "Fair Access Rule,"* https://www.sec.gov/tm/faq-regulation-ats-fair-access-rule (noting higher regulatory requirements above certain volume thresholds).

[291] Sec & Exch. Comm'n, *SEC Adopts Rules to Enhance Transparency and Oversight of Alternative Trading Systems*, https://www.sec.gov/news/press-release/2018-136; FINRA, *Where Do Stocks Trade*, https://www.finra.org/investors/insights/where-do-stocks-trade. On the registration requirements for exchanges, Exchange Act § 6(a), 15 U.S.C. § 78f(b) (2000).

[292] 17 C.F.R. § 242.301(b)(7).

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venue adhere to and push for basic standards of good conduct, disclosure, customer oversight, and customer protection. Digital asset ATS might seek to report their compliance to regulators and to an SRO. If structures seek to be highly decentralized, policymakers might consider allowing for compliance demands to be outsourced to an SRO altogether, such that an ATS relies entirely on a third party to perform such checks as vetting customers or monitoring for risks of price manipulation.

In the near-term, the goal lies in quickly enabling the creation of an SRO-based regime for exchanges transacting in digital assets. For Congress or specific agencies charged with regulation, developing detailed organizational, operational, and customer conduct standards might well take time. However, the creation of an SRO mandate offers a number of benefits, even where the full public regulatory framework remains under negotiation.[293]

First, it would require public regulators to create a specific regime for determining whether a venue can credibly function as an SRO. This requires specifying core organizational attributes that a crypto-exchange ought to have in order to properly regulate, monitor, and discipline. While far from the complete set of public regulations that ordinarily govern traditional exchanges, the fact of reviewing an exchange's competence for oversight can focus in on criteria that have the effect of improving the integrity and governance of the exchange firm. For example, an exchange needs to have a competent board, legal and compliance personnel, as well as auditors and accounting experts. Such staffing is foundational for the logistical task of oversight, enabling the exchange to engage in and implement decision-making about proper systems and controls for deciding who can use the venue, setting standards of conduct, verifying the character and solvency of those entering the venue, and assuring that they can transact legally and safely (e.g., no wash trading). Where an exchange performs a token listing function, personnel would have to include those with specific expertise in vetting the technological quality of coins and their underlying blockchains.[294] Cybersecurity risks must be contained. Those knowledgeable about issues of crypto-asset custody would be employed.

Regulating the marketplace demands that exchanges must maintain surveillance functions, systems for keeping records, and channels of communication with public regulators to supply information about potential breaches. That is, in order to show that it is a competent private overseer, exchanges must be capable of engaging with public regulators to implement new legislation, standards, and policies (e.g., in relation to money laundering), and identifying users to law enforcement who may not be compliant. As part of these functions, the exchange has to create groups of dedicated, high-quality, trained professionals that can enforce rules in a fair and trustworthy way. For

[293] As noted earlier, the draft Responsible Innovation Bill, proposed by Senators Lummis and Gillibrand, contains terms designed to create a self-regulatory regime for crypto-exchanges.

[294] Kiviat, Massari & Brummer, *supra* note [16] (highlighting the significance of technology as a key part of the risk matrix of crypto-related ventures).

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the exchange to be taken seriously as an SRO, its staff have to be trusted, sufficient for their processes to be understood as meaningful. This constrains the ability of exchanges to be *ad hoc*, casual, or inattentive to the task of whom they hire for the purposes of oversight and the processes implemented. Rather than a simple cost center, supervision would better represent a key business priority necessary to gain access to the digital access intermediation market – and to ensure compliance with public rulemaking.

That regulators enforce organizational changes within prospective exchanges is made clear by the vetting procedures put in place to admit a new venue into the national market for exchanges. Under Sections 6(b) and 19(a) of the Securities Exchange Act, the SEC has to satisfy itself that the venue is organized in such a way that it can meet the purposes of the Act and can extract compliance from exchange members and users.[295] An exchange must be institutionally able to demonstrate that its venue will operate in a way as to be free of fraud and misconduct, fair and equitable in its dealings, and showcase sound internal corporate governance. This means, for example, that the SEC will demand detailed information on how the venue is set up, its board of directors, and its key personnel, in order to assess their competence, independence and conformity with expected standards.[296] The board generally must include those that are independent of the exchange's membership and can represent investor interests.[297] Various committees can be required, such as those overseeing regulatory oversight within the venue. A regulatory oversight committee might further have to provide assurance the venue includes boards of personnel to hear appeals.[298]

These vetting procedures have the added benefit of creating greater transparency and opening otherwise opaque and complex venues to public scrutiny. An applicant exchange needs to furnish formal documentation, reveal sensitive information about its key personnel, as well as offer detailed insights about ownership structures. Whereas crypto-exchanges have struggled to shrug off worries about murkiness in their organizational set-up owing to the fact of being public and lightly regulated, the need to obtain SRO status will require revealing information to regulators and the public.[299] Further, the fact of applying to a regulator also necessitates a conversation to implement desirable changes. As typically happens in the context of new exchanges looking to gain SRO status, the SEC can engage with the applicant firm to proactively advise on optimal arrangements. Public comment on applications can also prompt

[295] 15 U.S.C. 78f(b); 15 U.S.C. 78s(a).

[296] 15 U.S.C. 78f(b)(3).

[297] *See e.g.*, Sec & Exch. Comm'n, *In the Matter of the Application of Investors' Exchange, LLC for Registration as a National Securities Exchange*, Release No. 34-78101, 4-7 (Jun. 17, 2016).

[298] Sec & Exch. Comm'n, *In the Matter of the Application of Investors' Exchange, supra* note [297], 13-15.

[299] Sec & Exch. Comm'n, *In the Matter of the Application of Investors' Exchange, supra* note [297], 16-18.

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applicants to consider changes or to justify internal arrangements as being compliant with the demands of the Securities Exchange Act.[300]

Secondly, the need to be designated by regulators as a major digital asset intermediary would likely motivate crypto-exchanges to preemptively reform their internal organization. Before a venue makes its application, it would likely consider it prudent to appear before a regulator with its house in good order. This ought to prompt investment in high-quality reform of organizational structures that may not pass review or might appear publicly damaging – such as an ineffectual board, a thinly staffed legal and compliance department, sloppy recordkeeping, susceptibility to hacks, or a non-cooperative attitude toward peers. In other words, the presence of the process itself should offer positive externalities by generating soft pressures to elevate the standard of governance well before the venue decides to put itself up for designation.

While not an exact comparison, this kind of anticipatory sprucing has long been evident in the case of companies contemplating going public. Start-ups or relatively young companies tend to lack the usual trappings of high-end corporate governance. As Elizabeth Pollman notes, start-up organizational culture often results in a complex mix of competing interests between founders, investors, managers, and employees. This can give rise to costly tensions and monitoring failures and evidenced in sub-optimal behaviors.[301] Scant attention might be paid to incorporating standardized accounting and auditing practices. Limited public scrutiny results in personnel preferences that do not fill important roles (like a general counsel).[302] As David Larcker & Bryan Tayan show in their study, however, companies start taking steps to transform their internal structures many years ahead of a potential public debut. Believing in a future offering causes firms to ramp-up hiring, switching to standardized auditing and accounting practices, and changing board composition.[303]

In other words, the hard deadline of a regulatorily and reputational significant event like an IPO causes companies to modify their behavior to fall more closely in line with expected standards of robust governance and risk management. For ATS venues aspiring to designation as an SRO, then, a set of standards can function as milestones that influence behavior long in advance of an actual application. This can result in gradual benefits for the marketplace where the presence of an SRO designation regime and its requirements fosters improvements in governance even for those companies that are not yet ready to formally apply.

Thirdly, SRO designation also requires public regulators to monitor venues for ongoing compliance. This means rebuking or punishing those that

[300] Sec & Exch. Comm'n, Comments on Investors' Exchange LLC; *Notice of Filing of Application, as Amended, for Registration as a National Securities Exchange under Section 6 of the Securities Exchange Act of 1934*, Release No. 34-75925, https://www.sec.gov/comments/10-222/10-222.shtml.

[301] Elizabeth Pollman, *Startup Governance,* 168 U. PA. L. REV. 155 (2019).

[302] David Larcker & Bryan Tayan, *Scaling Up: The Implementation of Corporate Governance in Pre-IPO Companies*, Stanford University Graduate School of Business Research Paper No. 19-11 (Nov. 30, 2018).

[303] Larcker & Tayan, *supra* note [302].

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fail to perform proper oversight. As detailed in Part II, the SEC has acted against venues like the NYSE or Nasdaq for various shortcomings in discharging their duties as SROs. Beyond the fact of sanction, monitoring can also entail approving rule changes, product launches, and providing input into internal changes that an SRO may contemplate that may impact its ability to perform its role. For example, for traditional marketplaces, amendments to exchange rules and changes to their trading processes are filed with the SEC for approval and subject to public comment. For instance, in late 2022, both the NYSE and Nasdaq sought approval for rules changes to enable them to provide new forms of listing (e.g., a direct listing).[304]

This kind of monitoring for crypto-exchanges will provide gains that are presently elusive. Importantly, as detailed in Part III, crypto-venues engage their customers and contractually bind them to terms of service. SRO status and regulator monitoring for compliance can mean that terms of service are subjected to thoroughgoing public scrutiny as these are modified and updated. For customers this process can offer an opportunity to read terms and comment. It can offer clarity on major risks that may be confronting them, such as the terms on which their assets are held, allowing them to choose the venue that is most favorable to their preferences. Further, it provide authorities with a continual understanding of the regulatory environment being created by an exchange. Where certain customers might get preferential terms (e.g., insiders, or large institutions), there may be pushback to rectify potential disparities in fair access. An overly cumbersome dispute resolution process may be considered unworkable within an SRO model that seeks to create high-quality due process and ensure that a venue does hold itself out as accountable.

One useful hypothetical here would be to consider whether qualifying criteria imposed by a public mandate for exchange self-regulation might have prevented FTX from engaging in the kinds of abuses that resulted in its customers facing $8 billion in losses. Hindsight, of course, offers a perfect view. Further, the scale of FTX's recklessness and misconduct was overlooked across multiple quarters.[305] Nevertheless, it is reasonable to suggest that some of the most dubious practices may have at least been checked by a careful vetting process to designate the exchange as an SRO, as well as from the exercise of the exchange's SRO's functions. For a start, FTX has emerged as a poster child for a deeply flawed and broken culture of corporate governance and risk management. Its Chapter 11 bankruptcy filing pointed to a company lacking almost all controls: no overarching board of directors, unqualified senior personnel, nominal attention to legal and compliance, completely botched

[304] Ropes & Gray, *SEC Approves NYSE's Proposal to Facilitate Primary Direct Listings by Modifying Pricing Limitations; Follows Approval of Nasdaq Proposal* (Dec. 19, 2022), https://www.ropesgray.com/en/newsroom/alerts/2022/december/sec-approves-nyses-proposal-to-facilitate-primary-direct-listings-by-modifying-pricing-limitations.

[305] *See e.g.*, Layan Odeh, *Investor Studied Crypto for Years, Then Missed FTX's Red Flags*, BLOOMBERG, NOV. 19, 2022; Sarah McBride, *Sequoia Capital Regrets Backing FTX But Defends Vetting Process,* BLOOMBERG, NOV. 22, 2022.

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recordkeeping, few internal walls between various parts of the internal organization, and an organizational structure that was opaque, highly complex and interwoven internationally across 100+ entities.[306]

The need to obtain designation as an SRO might have helped to address some of the problems. For example, FTX would have had to show that it was organized in such a way as to perform oversight. This would, almost certainly, require the need for a robust, qualified, and representative board. Recordkeeping and attention to customer activities and assets would be checked, alongside the presence of staff capable of overseeing customers and their assets. Audited financials would be required to demonstrate the venue's solvency, its financial good standing, and the health of its balance sheet. Even if such checks were performed only with respect to FTX.US – the U.S. arm of the Bahamian-based exchange – a regulator would still have to examine its overall ownership structure, the key personnel in charge, and the relationships between different divisions (e.g., the role of Alameda and the exchange). Further, the requirement for formal processes would likely create the need for truthfulness, legal opinions from expert counsel, and liability in case of lies or misstatements. While there is no guarantee that SRO status and the designation's need for regulatory review would necessarily have prevented the FTX collapse, it might at least have made it less calamitous – by the threat of regulatory monitoring, the need to account of organizational quality, monitoring of Alameda's role and its own financials, as well as safeguards on customer interests. FTX might still have fallen – Alameda taking big bets and failing, triggering mass customer withdrawals – but its fall may have implicated losses of FTX's own money, rather than that of customers.

3. Incentivizing Exchanges to Invest in Oversight

The major question for any SRO regime is why a crypto-venue would want to join it. It might still intermediate digital assets as an ATS-equivalent. Policy would, therefore, have offer some "carrots" that could make assuming regulatory responsibility economically worthwhile for crypto-exchanges.

First, policy might consider restricting the volume and growth potential of an ATS transacting in digital assets. In other words, just as a securities ATS designation places strictures on those hosting larger volumes, a digital assets ATS that transacts above a certain dollar volume of digital assets should have to seek out an SRO designation. This design would allow smaller venues to keep operating, subjecting them to limited responsibilities for oversight as long as they stay small. As they acquire volume and customers, and widen their

[306] FTX Trading, Voluntary Petition for Non-Individuals Filing for Bankruptcy, Case 22-11068-JTD (Nov. 11, 2022); In Re FTX Trading Ltd., Declaration of John Ray III in Support of Chapter 11 Petitions and First Day Pleadings, Case No. 22-11068 (JTD) (Nov. 17, 2022).

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footprint within the digital assets market, they become better placed to exercise meaningful oversight, and fall under the requirement to seek out an SRO designation. While this policy might operate in a harder way than a carrot, it does recognize those firms that are succeeding and acquiring market share. It includes them within the fold of those firms that are considered sufficiently robust commercially as to play their part in protecting the market.

Secondly, SRO status can come with some real-world positive pay-offs. An exchange that is recognized as having the size and stature to register as an SRO can end up perceived as acquiring legitimacy, and worthy of trust and confidence. This can bring customers to the venue, including those institutions that demand a venue capable of offering trust. With a boost in customers, it has scope to develop its differing kinds of business, such as a listing business, that can further broaden the impact of the venue on the marketplace. The move from an ATS to an SRO, with vetted governance, risk management, and compliance, can help reduce its overall cost of capital. As a safer, publicly accredited venue, performing a regulatory function, it may find it easier to raise funds from new investors, borrow more easily, and encounter an easier pathway while procuring licenses for expanded business opportunities. Greater transparency, sounder governance, reliable financials, and the potential of a broadened customer base, can offer comfort to investors performing due diligence, and limit the perceptions of riskiness attaching to the venue.

Thirdly, a possible innovation for digital asset markets lies in the creation of a national market for trading in crypto-assets, where SROs constitute the direct participants. As in equity markets, orders for particular coins would route to those venues that are offering them at the best price. This model for market design has been seen as important in building a liquid, competitive, and efficient marketplace for equity.[307] It may also be able to provide a structure for digital assets to similarly trade in a more liquid, competitive, and efficient marketplace, enabling customers to access a deeper float/pool of assets than they might get just on one venue, thereby benefiting market quality not just in cryptocurrencies like Bitcoin or Ether, but also digital assets with much lower liquidity. To be sure, this model would need some thoughtful engineering in the context of digital assets. For example, crypto-exchanges tend to custody customer crypto in proprietary wallets, creating some hurdles for transactions by a customer of one exchange to see rapid settlement when a trade executes on another. That said, a national market for digital assets can achieve two gains for venues seeking an SRO status. One, it can create pathways for exchanges to compete for a theoretically much bigger market. Two, the fact of being more closely connected to one another, pooling trading information, and being able to route trades to another venue, can foster pathways for greater cooperation

[307] Regulation National Market System Rule 611, Order Protection Rule, 17 CFR 242.611 (2005). On discussion of this model, Yesha Yadav, *Oversight Failure in Securities Markets*, 104 CORNELL L. REV. 1799 (2020) (detailing the tension between competition and oversight, as well as analyzing the National Market System framework).

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and collaboration. Where venues are sharing information more readily, as well as taking direct risks on one another, they are likely to have economic skin-in-the-game to oversee shared risks, monitor one another, as well as gain a stake in preserving the integrity of the market from which they make their profit. By allowing SROs to be the direct participants in this national digital asset market, they acquire the potential to make commercial gains as well as lower their costs of oversight through the creation of a repeat game with other SROs that are in a similarly situated position.

Finally, policymakers might consider incentivizing SRO uptake by giving SROs alone the chance to offer certain kinds of commercial products. As detailed in Part III, crypto-exchanges have evolved using a business model that relies on offering a suite of products to customers. These can include credit products like loans or credit cards, or various crypto-specific offerings that deploy customer crypto holdings as the basis for reward-based programs.[308] Certain such offerings can be extremely lucrative for venues. In the case of Coinbase, for example, "blockchain-based" rewards programs, which involve returns from practices like staking, raised around $63 million in revenue, around 11% of Coinbase's total net revenue. As the SEC announced a potential crackdown on staking in February 2023, Coinbase's share price tumbled around 22%.[309] Exchanges may also sell their data, premium custody services, advisory products, and so on.

Regulators might offer SROs wider latitude than ATS to offer commercial products to customers. In other words, the fact of being an SRO permits the venue to offer a range of commercial products to customers. By having access to an otherwise restricted spectrum of commercial opportunities than what an ATS can, the SRO mandate comes with positive cost-benefit trade-offs. To be sure, this model has its own considerable risks. Deep commercial relationships between a venue and its customers generate conflicts of interests that can limit the willingness of the venue to exercise discipline.[310] But this design offers a pathway to motivate platforms to plan and prepare for seeking out SRO designation, invest in the process, and deploy resources in the discharge of their duties. For venues that are poor overseers, the cost of breach has to hurt an exchange for it to matter. In a framework where rescinding SRO status, with the resulting loss of commerce this would entail, can operate as a stick, exchanges will also see a tempting carrot to focus on maintaining this accreditation and exercising sound private oversight.

[308] But see the SEC's recent crackdown on staking on exchanges, Andrew Ross Sorkin et al., *The S.E.C. Signals a Crackdown on Another Crypto Practice*, N.Y. TIMES, FEB. 10, 2023.

[309] Adam Morgan McCarthy, *Staking Revenues Under Threat as Coinbase Shares Tumble 22%*, THE BLOCK, FEB. 10, 2023.

[310] *See e.g.*, O'Hara & Macey, *supra* note [49].

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B. Structural Reform and Conflicts of Interest

Cryptocurrency exchanges have tended to adopt a more centralizing model than their traditional counterparts.[311] As set out in Part II, traditional exchanges usually rely on a network of entities to ensure that trades can move reliably from order submission to settlement. Once a bargain concludes on an exchange, its settlement can involve a host of actors. The most consequential is the clearinghouse – standing in the middle between buyer and seller to effectively guarantee performance. Exchanges also rely on recordkeepers and custodians for securities, cash, and commodities.[312] For example, the NYSE's equities trading operations rely on the Depository Trust and Clearing Corporation (DTCC) for clearing and recordkeeping services. By dint of this separation between entities, risk is more dispersed. To be sure, it is shared and likely with some contagion potential, given the interdependencies. Exchanges and clearinghouses can also form part of the same corporate group. The CME's clearinghouse – CME Clearing – represents a vertical silo between the exchange and its post-trade clearing and settlement apparatus.[313] The NYSE was a key mover in the creation of the DTCC.[314] Its organization maintains a member-owner interest in the clearinghouse.[315]

In contrast to traditional exchanges, cryptocurrency exchanges have adopted a model that relies on internalizing multiple trading, post-trading, and commercial functions within the same or closely related entities. As exemplified by FTX, exchanges may act as brokers, funding margin loans. Some provide market making services using their own balance sheet or those of related firms (e.g., Alameda). Venues can include units designed to fund start-up crypto-ventures. A suite of commercial products provides a strategy by which to closely tie customers to an exchange (e.g., offering loans).

Deep centralization creates serious risk for conflicts of interest that can tarnish the delivery of oversight. As in the case of FTX, where Alameda traded on preferential terms, an exchange may be reluctant to crack down on its own interests.[316] It may hesitate before punishing big customers, such as major hedge funds.[317] The mixing up of risky businesses – like venture investing or market making – alongside the exchange's main customer-focused trading enterprise

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[311] Yadav, *Centralization Paradox*, *supra* note [41].

[312] Dan Awrey & Joshua Macey, *Open Access, Interoperability, and the DTCC's Unexpected Path to Monopoly*, Cornell Legal Studies Research Paper No. 21-20, 17-23 (Jul. 2021) (detailing the role of the clearinghouse and the origins of the DTCC).

[313] CME Group, *Clearing*, https://www.cmegroup.com/clearing.html.

[314] Awrey & Macey, *supra* note [312], 20-21; Paolo Saguato, supra note [167],612-614 (2017) (noting the importance of clearinghouse ownership as a source of risk and surveying models of ownership).

[315] Depository Trust Corporation, *DTC Member Directory*, https://www.dtcc.com/client-center/dtc-directories.

[316] *See e.g.*, O'Hara & Macey, *supra* note [49].

[317] Joshua Oliver & Sujeet Indap, *FTX Businesses Owe More than $3bn to Largest Creditors*, FIN. TIMES, NOV. 20, 2022.

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poses the risk that it fails to focus resources on oversight. Its solvency may also be jeopardized. As highlighted by the collapse of FTX, riskiness from one arm (e.g., Alameda) can create conditions that ultimately affect the trading business as a whole, and customer incentives to remain on the venue.

When developing a regime for an SRO, Congress and agency regulators will have to pay close attention to the unique business models inhabited by a crypto-exchange. For the purposes of creating SROs, conflicts of interest must be addressed. The question is how. A comprehensive regulatory regime will have to determine how a crypto-exchange ought to be organized, and the capital and conduct requirements. But a few strategies may be helpful in addressing conflicts, and the issue of a centralized corporate structure.

As a starting point, policymakers have to decide whether certain kinds of activities ought to be either prohibited, made very expensive, or highly circumscribed for an exchange. Those that implicate its balance sheet and raise the risk that its transactional interests take ultimate preference, create a dual risk. They can damage the venue's solvency as well as impact the perception that the exchange can ever be fair. Of special note is the practice of an exchange acting as a market maker. With FTX's Alameda exemplifying the problem of a venue collapsing owing to its resident market maker (and risk-seeking investment fund), while also enjoying premier treatment, these structures ought to attract special scrutiny. Traditional markets deploy professional market making firms to transact on the venue – rather than using their own balance sheet to do so.[318] As crypto-exchanges enter the regulatory perimeter, subjecting them to similar constraints applies valuable lessons learnt from the traditional market to crypto-trading.

In addition to market making, other kinds of centralized functions also raise powerful risks. Many crypto-exchanges have constituted hubs for funding various kinds of ventures. Recall that FTX ventures failed with a portfolio of investments worth over $5 billion. Other exchanges have suffered large losses on their investments, such as Binance that confronted a loss of over $1 billion when its equity in Terra/Luna was wiped out.[319] Alongside market making, exchanges operating as active venture capitalists raise the danger that the health of their balance sheet gets jeopardized. Further, exchanges may be more forgiving about how they treat self-funded crypto-ventures that decide to list on the exchange. Where an emerging company issues a new coin on an exchange, there may be a risk, both perceived and actual, that the exchange smooths its path to listing, as well as encourages liquidity in its trading. Within an SRO structure, forbidding such a practice or steeply increasing its cost can work to ensure exchanges avoid the conflicts of interest and default risks involved.

Short of outright bans for certain activities, another possible pathway lies in clearly and demonstrably separating the functions of different units

[318] NYSE, *The NYSE Market Model*, https://www.nyse.com/market-model.
[319] Elliott, *supra* note [283].

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within the exchange's operation. This entails requiring each arm of the exchange's different parts to be properly capitalized and governed with strict adherence to rules against sharing personnel, information, and space on the balance sheet. Each unit should operate as its own entity, preferably legally and financially separate. This kind of separation can help safeguard against information from one unit (e.g., trading) being used to support another (e.g., market making). Further, it might help to blunt the perception that an organization is endemically conflicted. In the case of traditional markets, for example, exchanges seek to ensure that their regulatory functions are clearly perceived as separate from their trading and commercial ones. NYSE, for example, has a separate unit – NYSE Regulation – designed specifically for discharging its responsibilities under the Exchange Act.[320]

Another option might look to transparency and disclosure as a way to blunt the potential for conflict and self-dealing by exchanges. To the extent that the crypto-market might be perceived as distinctive, lacking the usual mechanisms of generating liquidity, policy might, for example, consider relying on clear and detailed disclosure to customers and to regulators as a way to mitigate risks. This option offers a solution that is perhaps most accepting of the current features of crypto-exchange design, while seeking to warn users and policymakers of the risks. For example, ATS within equity markets are afforded some latitude in how they structure their internal markets, with the caveat that regulators and users be informed about the risks that they face. Some flexibility in establishing an ATS can mean that they might only open their doors to certain categories of traders and exclude others. Customers seeking out this environment might opt-into one dark pool over another. The organizational flexibility of such venues has been revealed in various enforcement actions arising from ATS' disclosure violations, where an ATS operator said one thing to its customers, but acted otherwise, in a way designed to preference its own bottom line or those of certain types of traders. Credit Suisse, for example, failed to tell its subscribers to its ATS – Crossfinder – that it preferred to always route their orders to Crossfinder platform, giving Credit Suisse's interests a leg-up even when better orders might be available elsewhere. It also failed to disclose that certain high-speed traders were present and active on the platform and were being alerted about subscriber orders.[321] ATS must regularly publish disclosures about their organization, publicly committing themselves to offering a particularized trading ecosystem to users.[322]

By itself, this disclosure based model offers a potentially lighter touch that deprioritizes structural reform as a way to deal with conflicts of interest within an SRO, or otherwise leaves organizational overhaul to more detailed

[320] NYSE, NYSE REGULATION, https://www.nyse.com/regulation.

[321] Keri Geiger & Sam Mamudi, *Barclays, Credit Suisse Agree to Dark Pools Settlements*, BLOOMBERG, Jan. 31, 2016.

[322] Sec & Exch. Comm'n, *SEC Adopts Rules to Enhance Transparency and Oversight of Alternative Trading Systems*, https://www.sec.gov/news/press-release/2018-136.

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legislation. To be sure, it may always be combined with other solutions, where detailed and usable disclosures accompany how different, separated parts of an exchange do business, and where public transparency generates accountability by making a venue responsible for misrepresentation.[323]

Clearly, no solution is perfect. Ideas set out here provide a first step in reflecting on steps that may be taken to put crypto-exchanges on a structural footing where they can succeed as effective private overseers. However, the problem of conflicts will always linger. Securities exchanges, for example, still confront conflicts of interest owing to the fact of being profit- seeking entities dependent on customers for business.[324] In the case of crypto-venues, these commercial pressures may well be even more powerful than those faced by traditional financial exchanges, since competition is fierce and international – and cooperation is made more difficult as a result. That being said, without reforming the unique structure of crypto-exchanges as part of any SRO mandate, conflicts and commercial pressures are likely to weigh down their chances of achieving stature and recognition as industry self-regulators.

In summary, this Part proposes a mandate for self-regulation in cryptocurrency markets. It shows that a public mandate is a necessary condition to force exchanges to assume the role of private regulators. By creating a cost to inaction, a mandate brings the private interests of exchanges more closely in line with the public good. Regulators too can gain by virtue of leveraging an engaged actor with unique expertise and considerable reach and market power. But while such a mandate is necessary, it is not sufficient to fully secure the cryptocurrency market. For one, it is not a substitute for thoroughgoing public regulation. Crucially, it leaves gaps. Exchanges balance a duty of oversight against their commercial, private interests. They also face a complex and opaque informational environment that forces high costs on careful monitors. Though imperfect, self-regulation provides an important first step – one with a long pedigree in finance – toward systematic oversight of an innovative, evolving, and often poorly understood technology gaining ever-deeper reach into the traditional financial system and real economy.

V. CONCLUSION

The cryptocurrency market has emerged with stunning speed as a source of structural transformation within capital markets. It has proven to be highly popular, also reaching young people and underrepresented minorities who have faced high hurdles when seeking to access more traditional markets.

[323] Brummer, *Disclosure, Dapps and DeFi*, supra note supra note [20], 2-4 (on the difficulties posed by dense disclosures and the importance of clarity and utility in disclosure models).

[324] For a detailed discussion of the problem in equity markets, Yadav, *Oversight Failure*, *supra* note [73]; Macey & O'Hara, *supra* note [49].

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Crypto has spawned a sophisticated market where digital assets and their blockchains underpin billions of dollars' worth of financial products and services. Yet despite its reputation for decentralization, cryptocurrency exchanges constitute a highly centralized mainstay within the marketplace. Critical for popularizing the technology, they have been essential in shaping user experiences and also the risks that can arise where risk, greed, and organizational neglect take hold. As public regulation works to take shape, this Article looks to a tried-and-tested solution as an important first and fast step toward placing the cryptocurrency industry on a surer, safer footing. Exchange self-regulation forces the industry's central players to take responsibility. It aims to harness their skill and resources toward the development of standards, norms, and a culture of compliance. As in mainstream financial markets, self-regulation is not a stand-in for public oversight. It is a complement. But with innovation moving at breakneck speed and generating enormous risks for everyday users, a mandate for industry self-regulation can at least help regulators keep pace. In this way, this Article seeks to highlight a regulatory opportunity – through which the self-interest of often highly risky exchange firms can be more closely aligned with serving the public good and bringing credibility to an otherwise shattered cryptocurrency market.

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